



'Widening' indicator

Leveraging the
potential for
inclusive European
research and
innovation

STUDY

Panel for the Future of Science and Technology



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'Widening' indicator

Leveraging the potential for inclusive European research and innovation

The EU's Horizon Europe framework programme promotes European innovation. The current Horizon Europe programme uses 'Widening' measures to target Member States, support their capacity to innovate, close gaps between EU countries and move towards performance convergence. This study provides a framework and associated indicator options for establishing a systemic 'Widening 2.0' methodology which could prove instrumental in discussing post-Horizon Europe 'widening' support measures. It is based on assessment of research and innovation indicators of targeted 'widening' countries, looking at the economic and research impact that Horizon Europe may have had between 2018 and 2025.

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List of Abbreviations

CV	Coefficient of variation
EC	European Commission
ECOA	European Court of Auditors
EIS	European Innovation Scoreboard
EIT	European Institute of Innovation & Technology
ERA	European Research Area
ERC	European Research Council
ERDF	European Regional Development Fund
FDI	Foreign direct investment
FP	Framework programme
GDP	Gross domestic product
GERD	Gross domestic expenditure on research and development
HEU	Horizon Europe
HEI	Higher education institution
ICT	Information and communication technology
IPR	Intellectual property rights
IT	Information technologies
JRC	Joint Research Centre
MSCA	Marie Skłodowska-Curie Actions
OECD	Organisation for Economic Co-Operation and Development
PIC	Potential for innovation capacity
PRC	Private for-profit entities (excluding higher or secondary education bodies)
PUB	Public bodies
REC	Research organisations
R&D	Research and development
R&D&I	Research, development and innovation
R&I	Research and innovation
RTD	(DG RTD) European Commission Directorate-General for Research and Innovation
S&E	Scientists and engineers
SII	Summary Innovation Index from the European Innovation Scoreboard
SME	Small and medium-sized enterprises
SR	Success rate in applications to Horizon Europe
STEM	Science, technology, engineering and math
STI	Science, technology and innovation
THE	Times Higher Education Ranking

Executive summary

Despite broad consensus that science, research, and innovation are vital for Europe's future prosperity and security, the EU's innovation system is failing to deliver. The gap with the United States is deepening and China has overtaken the EU-27, alongside weak convergence among Member States. Evidence shows that this lag stems less from public research and innovation (R&I) funding than from weak business sector innovation, with Europe struggling to translate strong science into globally successful firms – an increasingly serious weakness in a fast-moving, artificial intelligence (AI)-driven and geopolitically competitive innovation landscape. While boosting EU innovation capacity is now a central policy priority and debates focus on increasing public funding, a key unresolved challenge remains how to balance excellence-based funding with broader participation, given persistent disparities in research and development (R&D) and innovation performance across Member States.

The 'Widening participation and spreading excellence actions' introduced in the previous framework programmes have been supporting Member States in improving their capacities to adopt and adapt new technologies. These initiatives aimed to strengthen research institutions by facilitating networking, partnerships with leading institutions, and recruitment of skilled staff. Although it is premature to evaluate the full impact of these 'Widening measures', recent assessments show that these are likely to deliver some early evidence of success stories. However, there seem to be persistent barriers for participation of 'Widening' countries and impacts are highly heterogeneous, often depending on complementary national efforts. Genuine sustainable change tends to depend on R&I investment and reforms at national level, challenging the attraction of international researchers, and long-term sustainability through effects generated from research results.

The EU's next framework programme for research and innovation will continue to support 'Widening' measures aimed at reducing R&D performance gaps between Member States, (European Commission communication of 4 July 2025). According to the draft proposal, the 'Widening' programme should focus on delivering 'concrete measures' for building the R&D capacity of countries that have been lagging in terms of participation in Horizon Europe. More emphasis will be placed on incentivising them to catch up, e.g. by monitoring and recognising efforts in their commitment to using national strategies to benefit from additional EU support, tailoring access to 'Widening' funding schemes on this commitment and by using the European Semester process for monitoring. It could also involve a larger share of structural funds invested in R&I. Consideration is also given to splitting eligible countries into categories, depending on their position in their advance towards catching up.

The overall aim of this study is to provide a framework for establishing a systemic 'Widening 2.0' methodology, which will be instrumental when discussing 'Widening' support measures once the Horizon Europe framework programme ends. It discusses new or improved existing metrics which can be used for assessing which Member States to select for which support measures, considering both the needs, but also the capacity for effectively 'catching up'. Overall, the proposed framework allows for sound policy options when discussing new R&I support programme structures for 'Widening'. The report proposes options for possible different choices, in which case their respective strengths and weaknesses are discussed.

The study used the latest 2025 version of a broad tool for assessing a country's innovation capacity – the European Innovation Scoreboard (EIS) developed by the European Commission's Joint Research Centre (JRC). The EIS variables were used as indicators for the various dimensions composing a country's margin for innovation capacity building, complemented, where relevant, with other sources. The dimensions for 'Absorptive capacity', 'Access to technology', and 'Creative capacity' (multiple dimensions) allowed the classification of 'Widening' countries into five categories of 'Potential for Innovation Capacity' (PIC).

Two principles guided the underlying analysis. First, as the analysis has shown the importance of 'systemic' performance, countries need to perform well on all the prerequisites for innovation capacity building. Second, when defining benchmark levels for scores, the study considered the position of countries on their innovation development path, i.e. scores will be compared with the overall 'Widening' benchmark.

The analysis illustrates that addressing new 'Widening' measures to unleash the innovation capacity of lagging countries to catch up is a complex challenge that requires more than simply addressing participation in EU framework programmes. It requires considering the heterogeneity among 'Widening' countries in their position and potential for innovation capacity building. Continued 'Widening' support could use the proposed PIC classification to tailor 'Widening' instruments according to identified Member State bottlenecks and weak spots. Hence, the focus could be not only on supporting participation in framework programmes in general, but on supporting participation in those programmes/pillars, tackling a country's weaknesses, and which measures have the best scope for improving a country's PIC.

Table of contents

1. The background to the 'Widening' EU policy scene	1
2. This report: Developing a framework for a 'Widening' methodology.....	3
3. Trends in EU science, technology and innovation performance and cohesion	4
4. The 'Widening' programme in Horizon Europe	7
5. 'Widening' countries' participation in Horizon Europe	9
6. Drivers for innovation capacity and for capacity for catching up: A short review of the literature	14
7. Indicators for assessing the potential of countries to catch up on innovation capacity	16
7.1. Assessing the position of EU 'Widening' countries on innovation capacity	17
7.2. Assessing the potential of EU 'Widening' countries to catch up on innovation capacity__	19
7.2.1. ABSORPTIVE CAPACITY (ABS CAP).....	19
7.2.2. Science & Technology (S&T) ACCESS.....	21
7.2.3. QUALITY OF THE RESEARCH SYSTEM (RES SYS).....	24
7.2.4. INNOVATION ACTIVE BUSINESS SECTOR (INN ENT)	27
7.2.5. LINKAGES (LINK)	30
7.2.6. INTELLECTUAL PROPERTY (IP)	32
7.2.7. ACCESS TO FINANCE (FIN)	34
7.2.8. IMPACT (EFF).....	37
7.3. Context variables for innovation Capacity Building in 'Widening' countries	40
7.3.1. Government Deficit	40
7.3.2. Rule of Law	41
8. Comparing gaps and convergence on factors shaping the potential for innovation capacity building	42
9. An assessment of the potential of 'Widening' countries for innovation capacity building	43
10. Participation of 'Widening' countries in the EU's Framework Programme and their Potential for Innovation Capacity	46
11. 'Widening' index options.....	48
12. Concluding recommendations	49
13. Caveats and suggestions for further analysis	50

References	51
Annex 1: Times Higher Education Ranking	52
Annex 2: Estonia's innovation policy (based on Veugelers & Schweiger (2016))	54

List of figures

Figure 1 – Trends in EU27's Innovation Performance (SII score)	4
Figure 2a – Trends in EU27's Cohesion on Innovation Performance: CV	5
Figure 2b – Trends in EU27's Cohesion on Innovation Performance: gap	5
Figure 3 – Trends in Innovation Performance (SII score) of the 'Widening' countries	8
Figure 4 – Participation of 'Widening' countries in Horizon Europe Programme by Type of Applicant and by Pillar (2021–2024)	10
Figure 5 – Success Rates of 'Widening' countries in Horizon Europe Programme by Type of Applicant and by Pillar (2021–2024)	11
Figure 6 – Gaps of 'Widening' countries in Participation in Horizon Europe	12
Figure 7 – The position and trend in EU Member States Innovation Capacity (SII 2018–2025)	17
Figure 8 – 'Widening' countries gap with EU27 average on Absorptive Capacity (2018–2025)	20
Figure 9 – 'Widening' countries gap with EU27 average on S&T Access (2018–2025)	22
Figure 10 – 'Widening' countries Foreign Firms presence	22
Figure 11 – 'Widening' countries gap with EU27 average on Quality of the Research System (2018–2025)	25
Figure 12 – 'Widening' countries gap with EU27 average on Quality of Higher Education 2025	26
Figure 13 – 'Widening' countries gap with EU27 average on INN ENT (2018–2025)	28
Figure 14 – 'Widening' countries gap with EU27 average on Business R&D (2018–2025)	29
Figure 15 – 'Widening' countries gap with EU27 average on LINK (2018–2025)	31
Figure 16 – 'Widening' countries gap with EU27 average on IP (2018–2025)	33
Figure 17 – 'Widening' countries gap with EU27 average on FIN (2018–2025)	35
Figure 18 – 'Widening' countries gap with EU27 average on Public Finance for R&D (2018–2025)	36
Figure 19 – 'Widening' countries gap with EU27 average on 2.1.1 and 2.1.3 (2018–2025)	36
Figure 20 – 'Widening' countries gap with EU27 average on EFF (2018–2025)	38
Figure 21 – 'Widening' countries gap with EU27 average on PROD (2018–2025)	39
Figure 22 – 'Widening' countries Deficit (2018–2025)	40
Figure 23 – 'Widening' countries Gap with EU27 average on Rule of Law	41
Figure 24 – 'Widening' countries gap with EU27 average and within CV on factors for Innovation Capacity (2025)	42

List of tables

Table 1 – Trends in Participation Rates of 'Widening' countries in the EU Framework Programme	9
Table 2 – Participation of 'Widening' countries in Horizon Europe Programme (2021-2024)	9
Table 3 – Variation in Horizon Europe participation among 'Widening' countries	12
Table 4 – Evidence on convergence in innovation capacity within and between EU27 'Widening' countries	18
Table 5 – Evidence on position and convergence in absorptive capacity within and between EU27 'Widening' countries and EU27	19
Table 6 – Evidence on position and convergence in S&T Access within and between EU27 'Widening' countries and EU27	21
Table 7 – Evidence on position and convergence in Quality of the Research System within and between EU27 'Widening' countries and EU27	24
Table 8 – Evidence on position and convergence in Innovative Enterprises within and between EU27 'Widening' countries and EU27	27
Table 9 – Evidence on position and convergence in Linkages within and between EU27 'Widening' countries and EU27	30
Table 9 – Evidence on position and convergence in IP within and between EU27 'Widening' countries and EU27	32
Table 10 – Evidence on position and convergence in Access to Finance within and between EU27 'Widening' countries and EU27	34
Table 11 – Evidence on position and convergence in Impact within and between EU27 'Widening' countries and EU27	37
Table 12 – Position of 'Widening' countries on the factors shaping the potential to develop and catch up in innovation capacity	44
Table 13 – Assessing 'Widening' countries on their potential to develop and catch up in innovation capacity	45
Table 14 – Matching 'Widening' countries Position on Innovation Capacity with their participation in Horizon Europe	47

1. The background to the 'Widening' EU policy scene

There is no lack of reports these days arguing how important science, research and innovation are to take Europe into a prosperous, clean, safe and healthy future for its citizens. 'It is investment in research and innovation that will ensure Europe's (and its Member States') future security, strength and freedom and thus Europe's ability to successfully navigate in turbulent times' (Draghi (2024)¹).

Yet, the EU's innovation system is currently not in shape to deliver. The evidence showing how big the gap has become between EU's research and innovation system with the US and China is not new (e.g. Veugelers (2024)²), but the Draghi (2024) report proved to be a loud wake-up call. The innovation gap relative to the US is 'Widening' rather than shrinking, and China has managed to overtake the EU27. Together with the lacking EU catching up worldwide, there has been very slow convergence intra EU27, hampering EU's overall innovation performance.

A deeper dive into the evidence reminds us that the major source of EU's persistent lagging behind and failure to catch up with the US is not so much in its public R&I funding, but in its business sector R&I. While Europe may have nurtured pockets of great science, it typically succeeds less in turning them into innovative corporate successes, compared to the US, and more recently China. This has been a long-diagnosed challenge for the EU, which policy does not seem able to address. This missing dynamic business innovation has become even more critical in the current fast changing and highly competitive global AI powered innovation environment. In addition, the global innovation scene is increasingly being characterised by a race between US and China for technology sovereignty and dominance, protected by domestic fences.

Powering EU's innovation capacity has become central in current EU policy priority discussions. Reports of various stakeholders preparing their position on the upcoming new EU budget for Research, FP10 (e.g. the Heitor (2024)³ report), complemented the momentum. Most of the policy discussion calls for an increase in public funding, although there is no consensus between Member States on whether and how to support R&D through EU-level instruments or through coordination of instruments at the level of Member States.

Considering evidence indicating weak or no convergence in R&D and innovative performance across Member States, the question is how to reconcile broader participation in the public and private R&D&I ecosystems in the EU, with excellence-based public and private selection. Differences in R&D&I performance create a divide that affects participation in public programmes such as the Framework Programmes (FPs) where funding is awarded based on excellence. Lower participation further limits underperforming countries' ability to improve their research and innovation systems.

The 'Widening' Participation and Spreading Excellence actions' introduced in the current FP Horizon Europe (HEU), is targeted at supporting Member States in improving their capacities to adopt and adapt new technologies. These initiatives aimed to strengthen research institutions by facilitating networking, partnerships with leading institutions, and recruitment of skilled staff.

Although it is premature to evaluate the full impact of these 'Widening' measures', several recent studies, e.g. by the EC itself and the European Court of Auditors (ECA) have assessed the changes brought by the initiative (ECA, 2022)⁴. Although this evidence shows that the measures seem to

¹ European Commission (2024), 'Draghi report' https://commission.europa.eu/topics/competitiveness/draghi-report_en

² Veugelers R (2024) An Innovation-Based Industrial Policy for the EU. *Intereconomics* 59(5): 254–261 <https://reference-global.com/download/article/10.2478/ie-2024-0052.pdf>

³ European Commission (2024), 'Heitor report' <https://data.europa.eu/doi/10.2777/9106236>

⁴ [Special Report of the European Court of Auditors No 15/2022: 'Measures to widen participation in Horizon 2020 were well designed but sustainable change will mostly depend on efforts by national authorities'](#).

deliver some early evidence of success stories, there seem to be persistent barriers for participation of 'Widening' countries and impact highly heterogeneous, often depending on complementary national efforts. Genuine sustainable change seems to depend to a large degree on R&I investments and reforms at national level, challenging the attraction of international researchers, and long-term sustainability through effects generated from research results.

The EU's next Framework Programme for research and innovation, FP10, will continue to support 'Widening' measures aimed at reducing R&D performance gap between Member States, (EC July 4 communication)⁵. According to the draft proposal, the 'Widening' programme should focus on delivering concrete measures for building the R&D capacity of countries that have been lagging in terms of participation in Horizon Europe. More emphasis will be placed on incentivising them to catch up, e.g. by monitoring and recognising efforts in their commitment to using national strategies to benefit from additional EU support, e.g., by tailoring access to 'Widening' funding schemes on this commitment and by using the European Semester process for monitoring. It could also involve a larger share of structural funds invested in research and innovation. It also considers splitting eligible countries into categories, depending on their catching up position.

⁵ COM (2025) 543 final 2025/0543 (COD) <https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52025PC0543>

2. This report: Developing a framework for a 'Widening' methodology

Evaluating and monitoring of 'Widening' measures requires an effects-based approach entailing *ex-ante* and *ex-post* micro and macro assessments of long- and short-run impacts of specific instruments and of the FP portfolio in general. To what extent does 'Widening' Participation and Spreading Excellence' unleash the innovation capacity of lagging Member States? The relevant key performance indicator to use is the contribution of the portfolio of instruments to nurture the next scientific and technology ideas that will boost business innovation and growth.

In the light of the ongoing discussions on the role of 'Widening' measures' in the next FP10 (2028–2034) or its alternatives (e.g. [European Competitiveness Fund](https://commission.europa.eu/publications/european-competitiveness-fund_en))⁶, it is important to reassess and update existing metrics defining the distribution of these measures in the R&I support programme after HEU.

The overall aim of this study is to provide a framework for establishing a systemic 'Widening' 2.0' methodology, which will be instrumental when discussing 'Widening' support measures after HEU. It will discuss new or improved existing metrics ('Widening' 2.0' indicator) used for assessing which Member States to select for which 'Widening' support measures, considering both the need for 'Widening' support, but also the capacity for effectively 'catching up'.

The framework for such a systemic 'Widening' 2.0' methodology will focus on indicators that are relevant, measurable, achievable, and time-bound:

- The methodology should be multidimensional, building on various metrics involving Science, Research, Technology, Innovation, Business Structure, Productivity.
- The methodology will be relevant for an *ex-ante* assessment of strength and weaknesses in R&D&I of Member States as well as for an assessment of the capacity for catching-up through 'Widening' (openness, absorptive capacity, creative capacity). Consequently, metrics should go beyond R&I investment level as % of GDP (total, public, private) to also include metrics such as number of people and skills trained for R&D&I, diffusion/connectedness measures such as metrics of researchers' mobility and cooperation.
- The framework and associated indicator options will consider the dynamic dimension of metrics that are used to assess initial positions, as well as trends and patterns of growth/catching up.
- The framework should consider (trends in) Horizon participation metrics.
- The framework will critically assess its composite dimensions, with appropriate differentiation of dimensions when needed.

Overall, the proposed framework will allow for sound policy options when discussing new R&I support programme structures for 'Widening'. The report will propose options for possible different choices, in which case their respective strengths and weaknesses will be discussed. The proposed framework will also be useful for future bigger in-depth studies for evaluating R&D&I capacity development across Member States and time.

⁶ https://commission.europa.eu/publications/european-competitiveness-fund_en

3. Trends in EU science, technology and innovation performance and cohesion

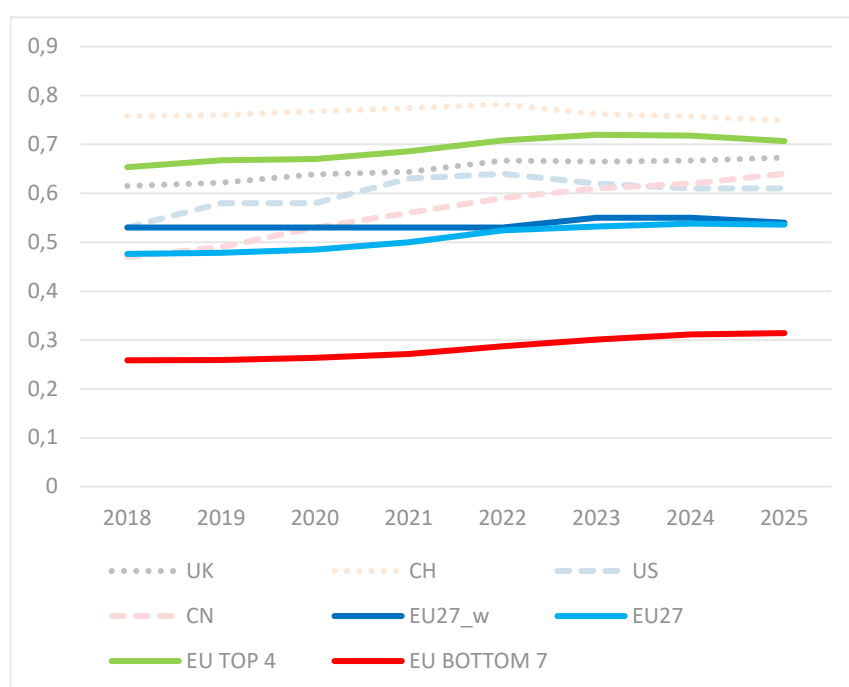
The **innovation capacity** of nations measures their ability to generate new ideas and to translate them into economic growth and prosperity (Furman et al, 2002)⁷. Because of differences in initial conditions and because of differences in how EU countries have sought to create innovation-based growth, we can expect substantial differences between European countries in terms of innovation capacity. We would however expect that the process of EU integration would allow lagging countries to catch up faster, pushing convergence within the EU in terms of innovation capacity, along with economic convergence.

A broad tool for assessing a country's innovation capacity is the European Innovation Scoreboard (EIS) developed by the European Commission's Joint Research Centre (JRC).⁸ The EIS uses multiple indicators (32 in recent editions), but the overall score for each country is represented by Summary Innovation Indicator (SII).

This section will use the latest 2025 version of SII scores to trace trends in EU-27's overall position in innovation capacity, as well as trends in the cohesion within the EU-27 on this indicator.

Overall EU Science, Technology and Innovation (STI) performance, as measured by the trends in its SII score relative to the rest of the world, is showing, in line with Draghi report et al, how EU is losing in relative position in innovative performance. US and China are all scoring consistently better than

Figure 1 – Trends in EU27's Innovation Performance (SII score)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

⁷ Furman JL, Porter ME, Stern S (2002) The determinants of national innovation capacity. *Research Policy* 31(6): 899-933

⁸ https://research-and-innovation.ec.europa.eu/statistics/performance-indicators/european-innovation-scoreboard_en

the EU-27. And the trend is not one of EU-27 consistently catching up. In contrast, and particularly relative to China, EU-27 is consistently losing ground.

Parallel with the EU27 losing grounds relative to US and China, there is a high and quite persistent variation in innovative performance among the Member States within EU27. To measure the variation in innovative performance, we use the Coefficient of Variation⁹ (CV), which provides a standardised indicator for variation comparable across series. The higher the CV score, the larger is the variation. A CV score that is declining over time signals a convergence (called 'sigma-convergence'), i.e. a reduction in the dispersion. As an alternative measure for cohesion, we also calculate the gaps in SII score between the EU27 average and two groups of Member States (above or below the EU average). Gaps declining over time signal a cohesion (called 'beta-convergence').

Figure 2a – Trends in EU27's Cohesion on Innovation Performance: 'sigma-convergence' view

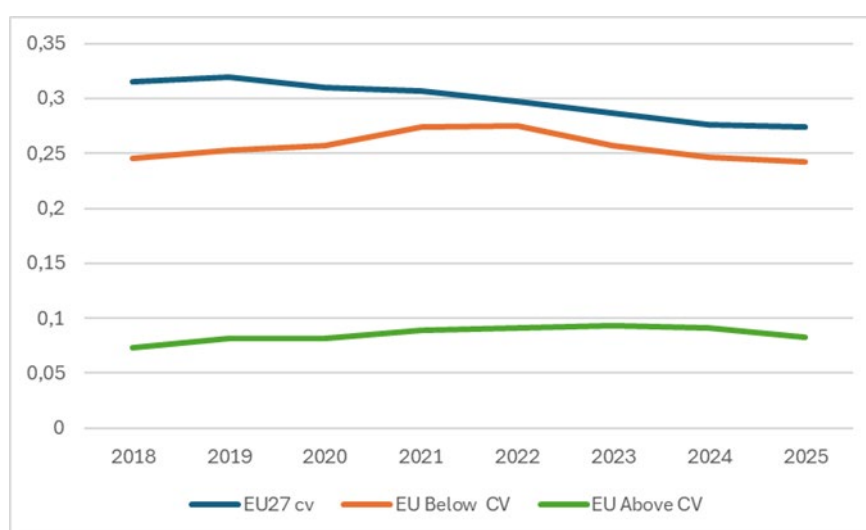
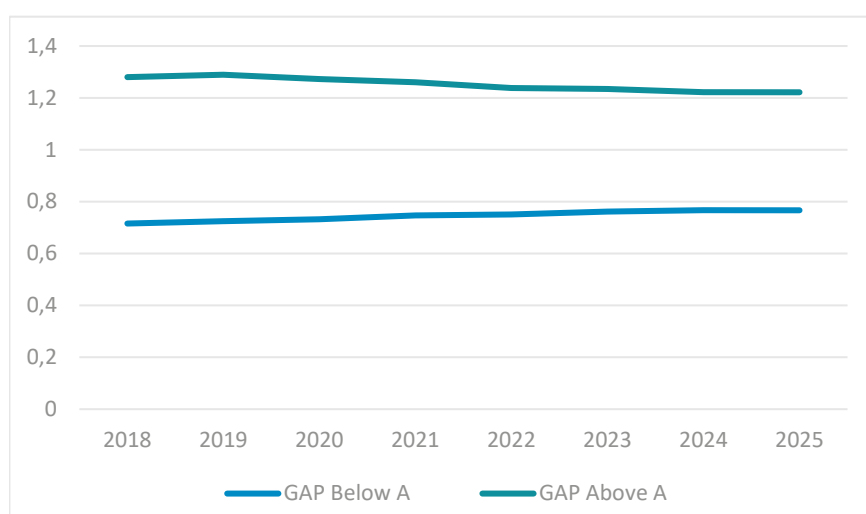


Figure 2b – Trends in EU27's Cohesion on Innovation Performance: 'beta-convergence' view



⁹ Coefficient of Variation (CV) is a statistical measure of the dispersion of a data set relative to its mean, calculated by dividing the standard deviation by the mean and expressed as a percentage.

The upper panel (Figure 2a) shows the Coefficient of Variation, measuring the dispersion within the EU-27. It shows that the variation has declined over time ('sigma-convergence'), but only slightly so. Similarly, the lower panel (Figure 2b) shows that the difference in SII performance relative to the EU-27 average (gap) between the countries above the EU average and the countries below the EU average has somewhat declined. A closing gap is another indicator of convergence ('beta-convergence'). However, this decline has been very minor. In particular, the gap of the countries below EU average has remained stable, reflecting little to no catch-up on average.

In addition, the upper panel (Figure 2a) also shows that the variation within the group of countries below the EU average (line denoted 'EU below CV') is substantial and significantly larger than the variation in the group of countries above the EU (line denoted 'EU above CV'). This variation among the EU 'laggards' has remained persistent over time and even went up a bit in the period 2020-2022.

Overall, the evidence confirms the difficult process of convergence and the persistent heterogeneity among the lagging countries within the EU-27. All this correlates with the persistent lagging behind of the EU-27 overall relative to the US and China. Although no proof of one 'causing' the other, these graphs suggest that the lack of a smooth cohesion process holds back overall EU's catching up in innovation performance relative to the US and China.

4. The 'Widening' programme in Horizon Europe

With its dedicated '*Widening participation and spreading excellence*' programme in its Framework Programme, the Commission has the objective to address the cohesion within the EU on innovative performance, by 'Widening' the participation in the Framework Programme of lagging countries and to improve their capacities in research and innovation, as well as to support the reforms of national policies, within an overall objective of strengthening the European Research Area.

Countries which are targeted 'Widening' countries', are less performing in terms of research and innovation with respect to the average EU 27. Within the Horizon Europe regulation 'Widening' countries are: Bulgaria (BG), Croatia (HR), Cyprus (CY), Czech Republic (CZ), Estonia (EE), Greece (EL), Hungary (HU), Latvia (LV), Lithuania (LT), Malta (MT), Poland (PL), Portugal (PT), Romania (RO), Slovakia (SK), Slovenia (SI) and all associated countries with equivalent characteristics in terms of R&I performance and the Outermost Regions.

The actions within this "Widening" objective are mainly *Coordination and Support Actions*, i.e. funding schemes sustaining activities that contribute to the objectives of Horizon Europe Programme, normally excluding research and innovation activities. All organisations eligible in Horizon Europe can participate in the 'Widening' actions but only organisations based in 'Widening' countries can participate as coordinators. Its main types of actions are:

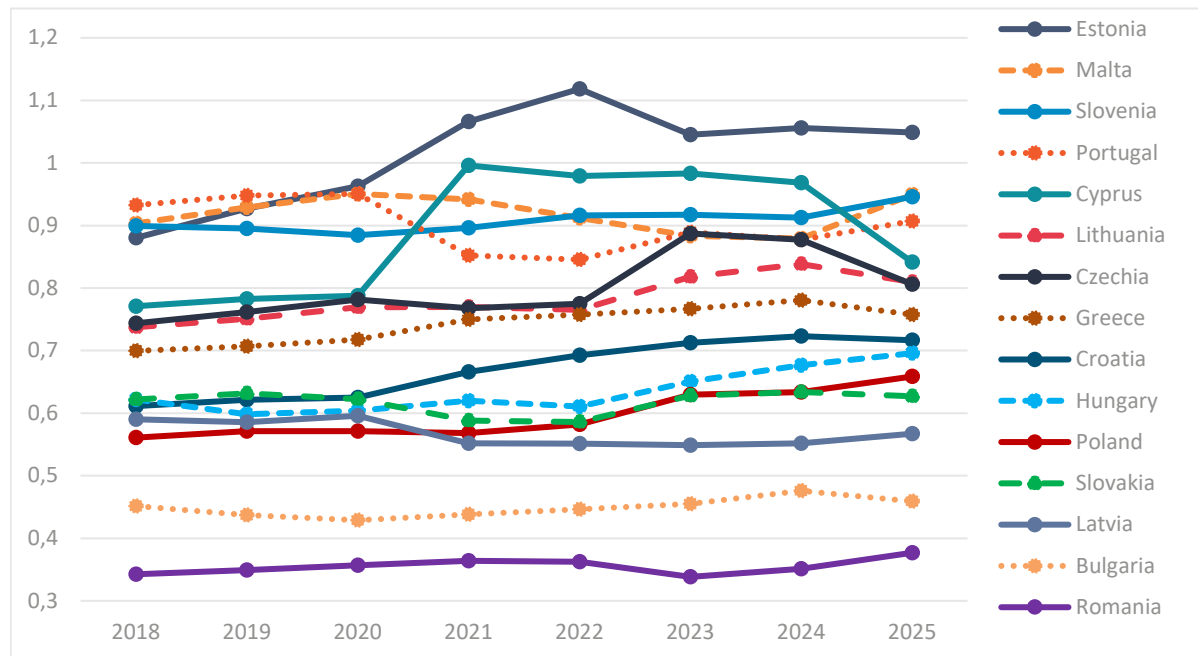
- **Teaming:** action of support to the creation of new centres of excellence or the upgrade of existing ones, located in 'Widening' countries, building on partnerships at European level. At least two parties must be involved in the action, a legal entity established in a 'Widening' country operating as coordinator and, a counterparty, a university or research organisation, internationally recognised for excellence in research and innovation, established in another EU Member State or Associated Country.
- **Twinning:** supports joint research projects, new or ongoing, aimed at strengthening scientific excellence and innovation capacity in a defined research area as well as the scientific quality of the partners involved in the collaborations. Under this action, a research institution located in a 'Widening' country act as coordinator in a partnership with at least two other internationally leading institutions/bodies located in other Member States or Associated Countries.
- **Pathways to Synergies:** This action supports beneficiaries of regional funding to increase cross-border collaborations and reduce access barriers to different funding systems.
- **Hop-on Facility;** This action aims to facilitate 'Widening' countries' research institutions joining already ongoing Horizon R&I projects.
- **ERA Fellowships:** ERA Fellowships are open to researchers moving to the EU from a third country or moving within the EU to a 'Widening' country.

All the 15 EU27 'Widening' Member States have scores below EU average on SII on average in the period 2018–2025, confirming their selection as 'Widening country'.

Most of them have made some progress in reducing their innovation gap (see Figure 3 below). This holds most notably for Estonia which has made most progress and even managed to score above the EU27 average since 2021. Portugal was the best performing 'Widening' country in 2018, but has struggled to reduce its remaining gap since, testimony of the challenge of building a long-term consistent catching up process.

The countries at the bottom have made little progress and continue to be persistent laggards, testimony of the challenge of catching up, particularly if the lags are considerable.

Figure 3 – Trends in Innovation Performance (SII score) of the 'Widening' Countries



Source: Own calculations based on [EC-JRC EIS2025](#) data.

The countries which are below 75% of the EU27 SII score (NB: a benchmark considered as a 'Widening' criterium in the next framework programme, cf. supra) are since 2022: HU, SK, HR, LV, PL, BG and RO.

In 2018, also CZ, LI, EL scored below 75%, but they have gradually made sufficient progress on SII to no longer be below 75%. While HU and PL have made some progress on reducing their gap, they nevertheless remain at below 75%. All this further illustrates the considerable heterogeneity in innovative performance within the current group of 15 'Widening' countries and the heterogeneity in catching up performance.

5. 'Widening' countries' participation in Horizon Europe

Overall, the participation of 'Widening' countries in the Framework Programme is substantially below the participation of non-'Widening' countries, again motivating their selection for "Widening".

But the gap in participation rates has been reduced in the current Horizon Europe programme, compared to the previous Horizon2020 ([EC-RTD HEU Mid-Term Evaluation](#), European Commission (2025)). The 'Widening' Member States have a higher share of participations in collaborative projects under Horizon Europe than previously under Horizon 2020, accompanied by a higher EU contribution. There is also a significant rise in the share of collaborative projects involving 'Widening' Member States – from 47% under Horizon 2020 to 58% under Horizon Europe.

Table 1 – Trends in Participation Rates of 'Widening' countries in the EU Framework Programme

	'Widening' countries		Non-'Widening' countries	
	Horizon Europe	Horizon2020	Horizon Europe	Horizon2020
Share of participation in collaborative projects	18%	15%	65%	73%
Share of EU funds in collaborative projects	15%	11%	77%	82%
Share of collaborative projects involving country group	58%	47%	97%	96%

Source: European Commission (2025), Interim Evaluation of the Horizon Europe Framework Programme for Research and Innovation (2021 – 2024)

In what follows we look in more detail into the participation of the 'Widening' countries in the most recent FP HEU interim evaluation by EC¹⁰.

Table 2 – Participation of 'Widening' countries in Horizon Europe Programme (2021-2024)

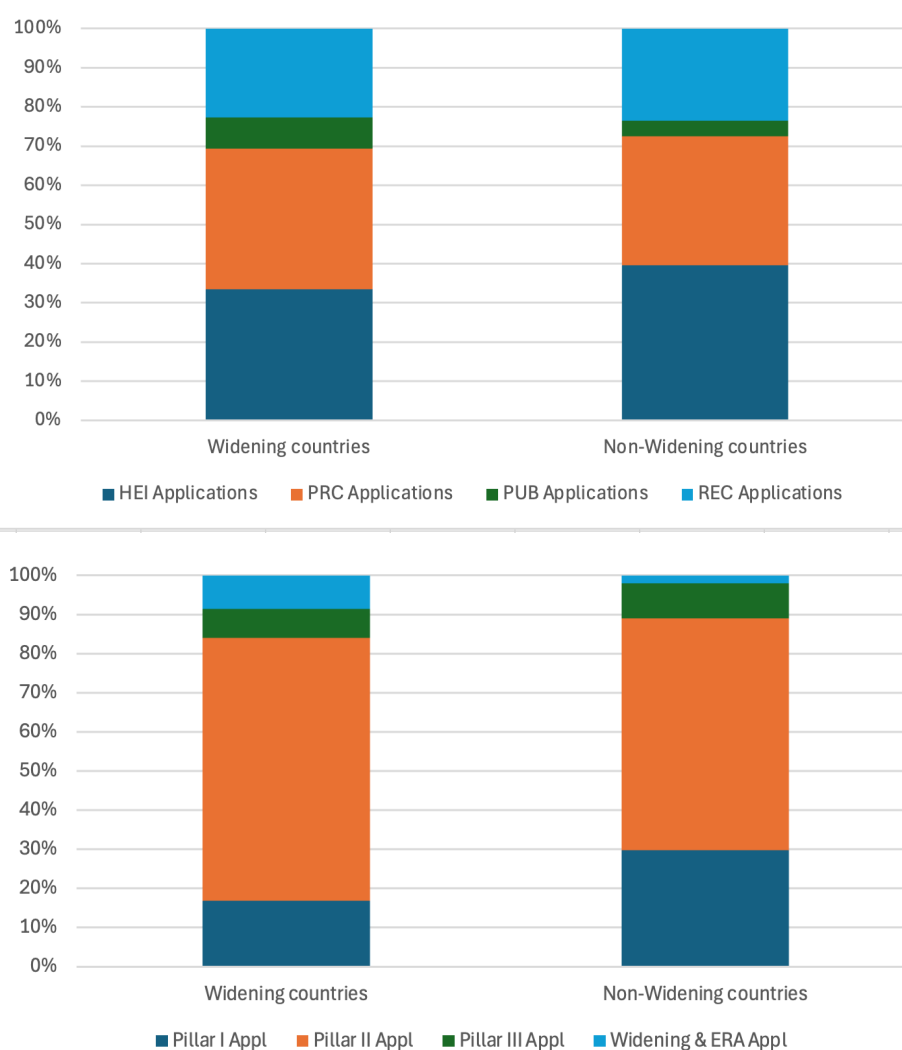
	EU27	'Widening' countries	Non-'Widening' countries
HEU Applications (2021-2024)	380 816	92 303 (24%)	288 513 (76%)
HEU Applications per n000 S&E	20	19	21
HEU Success Rate	22	19	23
HEU Participations (2021-2024)	85 351	17 910 (21%)	67 441 (79%)
HEU Participations per n000 S&E	4.5	3.7	4.9
HEU Investments mEUR	39 451	5 952 (15%)	33 498 (85%)
HEU Investment per mEUR GERD	10 707	17 490	10 017

Source: European Commission (2025), Interim Evaluation of the Horizon Europe Framework Programme for Research and Innovation (2021 – 2024). Legend: GERD (Gross domestic expenditure on Research and Development); per n000 S&E (per 1,000 scientists and engineers).

¹⁰ Interim Evaluation of the Horizon Europe Framework Programme for Research and Innovation (2021 – 2024), <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex:52025SC0110>

Overall, the 'Widening' countries represent approximately one-fifth of all participations in Horizon Europe during the period 2021-2024. When expressed relative to the number of active S&E

Figure 4 – Participation of 'Widening' Countries in Horizon Europe Programme by Type of Applicant and by Pillar (2021-2024)



Source: Own calculations based on European Commission (2025), Interim Evaluation of the Horizon Europe Framework Programme for Research and Innovation (2021 – 2024). Legend: HEI stands for Higher Education Institutions, PRC – Private-for-profit entities, PUB – Public bodies, REC – Research Organisations.

employees, the difference in participation with the non-'Widening' Member States is smaller, but nevertheless still substantially lower.

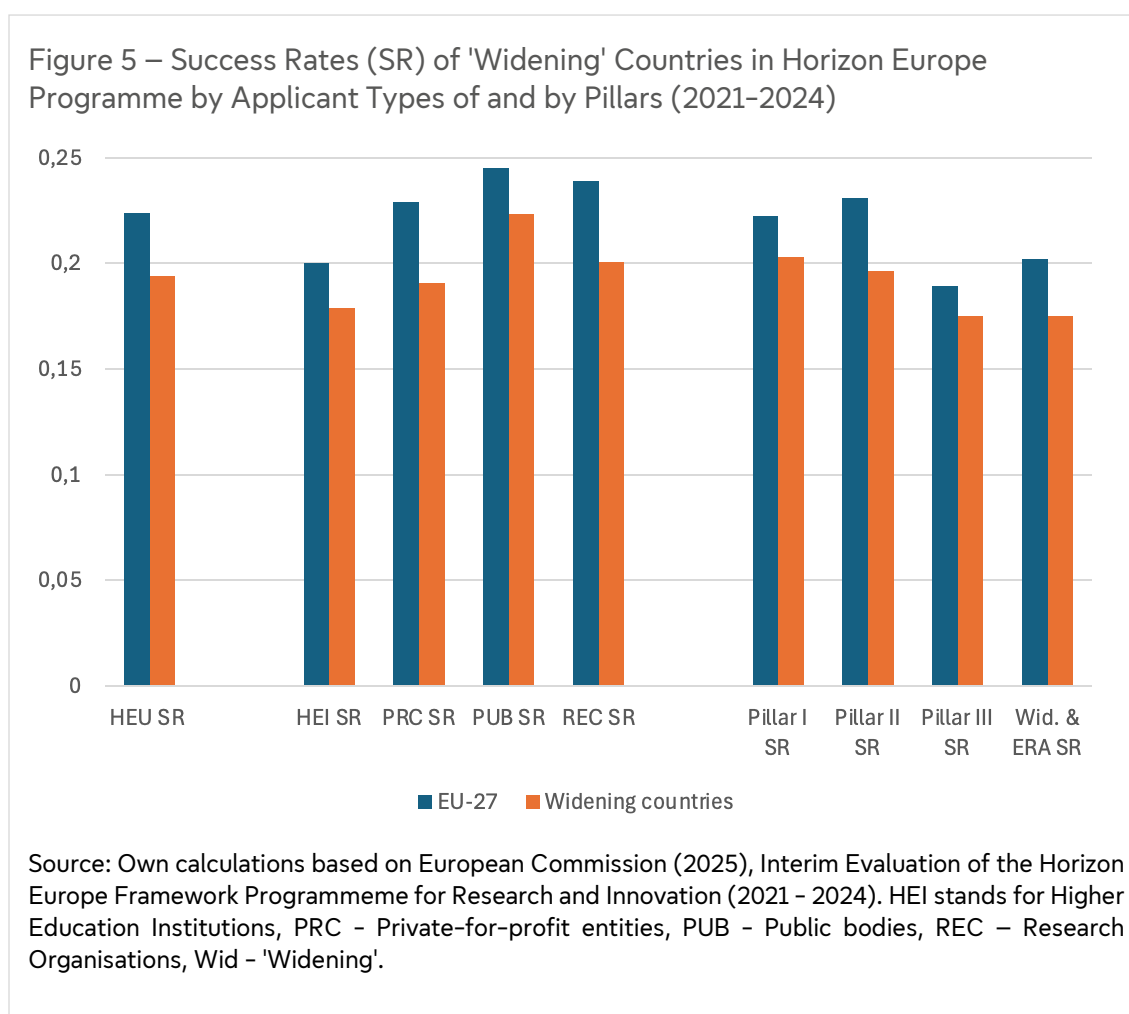
The lower participation rate of 'Widening' Member States is both in the number of applications as well as in their Success Rate (SR), the latter with a somewhat larger difference. When looking at the funds (investments) associated with the grants, the 'Widening' share is smaller than their participations, indicating that they are relatively more present in smaller grants. Figure 4 details below details SR according to HEU four Pillars: Pillar 1 (Excellent Science) for frontier research (European Research Council, Marie Skłodowska-Curie Actions), Pillar 2 (Global Challenges & European Industrial Competitiveness) tackling societal issues via clusters (Health, Digital, Climate, etc.), Pillar 3 (Innovative Europe) fostering market-creating innovation (European Innovation

Council, European Institute of Innovation & Technology), Pillar 4 cross-cutting pillar for 'Widening' Participation and strengthening the ERA.

When expressing the HEU funds received relative to the GERD of the involved countries, the ratio for 'Widening' countries is 75% higher than the EU27 average, expressing that the EU HEU funds compensate for the low national funding.

When looking within the Horizon Europe programme, the biggest pillar in terms of applications is Pillar II. The difference in application rate for 'Widening' countries is however largest for Pillar I (see Figure 4, upper panel). When looking at the type of applying entities, PRC (for-profit entities) are the largest group for 'Widening' countries (33%), while HEI (Higher Education) are the largest group for non-'Widening' countries (37%), but overall the differences are small.

The gap in success rate (SR) for 'Widening' countries often focuses on Pillar I (ERC & MSCA) and for HEI institutions, but the gap holds across Pillars and types as shown in Figure 5. It is even somewhat higher for the Pillar II and for for-profit organisations (PRC), suggesting that the focus of the 'Widening' programme should not be too narrowly on research and science, but broader across the whole science, technology and innovation spectrum.



When looking at the heterogeneity among 'Widening' countries with respect to HEU participation indicators, we see that there is not much difference in success rates across 'Widening' countries (see Figure 6 and Table 3 below). While on average the gap is 0.84, the 'Widening' country with the smallest gap, Slovenia, scores 0.93, while Romania has a gap-score of 0.75.

Table 3 – Variation in Horizon Europe participation among 'Widening' countries

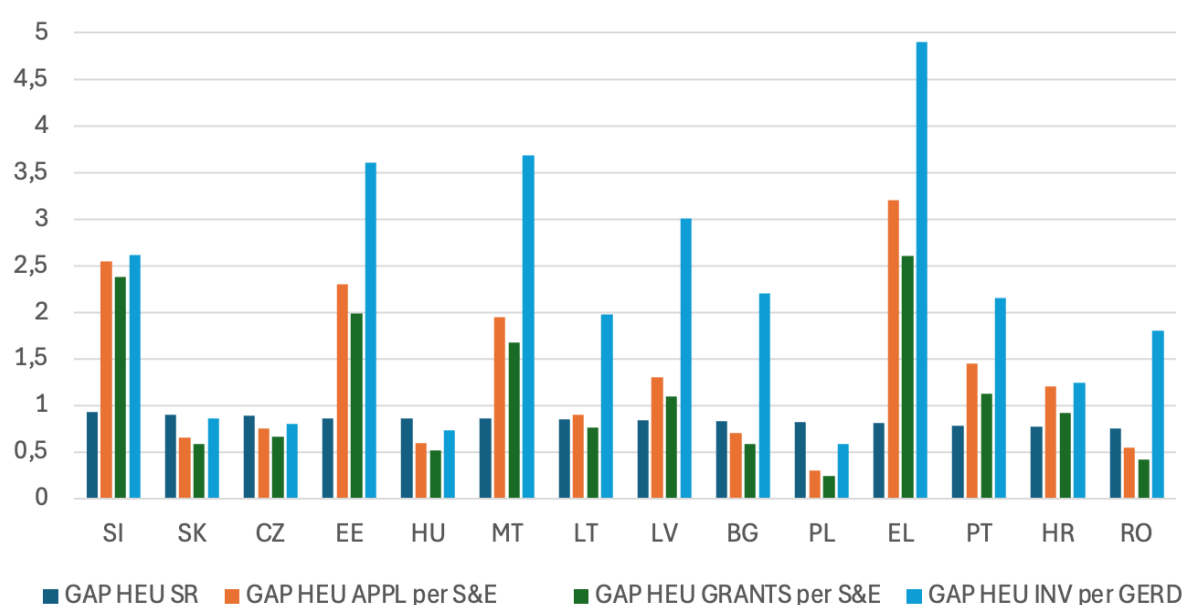
	HEU SR	HEU APPL per S&E	HEU GRANTS per S&E
CV 'WIDENING'	0,068	0,842	0,784

Source: Own calculations based on EC-RTD (2025), Interim Evaluation of the Horizon Europe Framework Programme for Research and Innovation (2021 – 2024)

However, once we look at the number of applications (per n000 S&E), the variance increases considerably. The success rate somewhat mitigates the differences in application behavior, as the variance in the grants (per n000 S&E) is somewhat smaller, but nevertheless still remains substantial. And finally, when looking at the funds/investment per GERD, the variance among 'Widening' Member States is substantially larger.

All this reflects that the major source of variance among 'Widening' countries with respect to participation in HEU rests in the local STI environment of the particular 'Widening' country (Figure

Figure 6 – Gaps of 'Widening' Countries in participation in Horizon Europe



NB: Cyprus is not reported as it is an outlier (HEU SR = 16; HEU APPL per S&E = 106; HEU INV per GERD = 17)

Source: Own calculations based on European Commission (2025), Interim Evaluation of the Horizon Europe

6).

We conclude this section with a short summary of the evaluations of the 'Widening' instruments in the Framework Programmes, as reported in the Mid-Term Evaluation of HEU Report from DG RTD (European Commission, 2025). The *European Court of Auditors*¹¹ has highlighted the difficulties faced by entities in 'Widening' countries in recruiting international staff due, for example, to the low attractiveness of a research career (and salary) in certain 'Widening' countries. Encouraging mobility

¹¹ [Special Report of the European Court of Auditors No 15/2022: 'Measures to widen participation in Horizon 2020 were well designed but sustainable change will mostly depend on efforts by national authorities'](#).

and new collaborations is particularly important, as researchers in 'Widening' countries are still less connected than those in non-'Widening' countries.

Stakeholders participating in the *survey* for the Mid-Term evaluation of HEU (European Commission, 2025) reported the difficulty to join collaborative projects and find consortia that would include them in running projects (like in the Hop-on Facility), due to lack of contacts. They also associated the reasons for low engagement from 'Widening' countries as largely systemic. They advocated to leverage budget from other EU programmes like the European Regional Development Fund (ERDF) and NextGeneration EU to structurally invest in R&I and reduce territorial unbalances. Unfortunately, the *ex-post* evaluation of Horizon Europe shows mixed findings on current synergies with the ERDF.

6. Drivers for innovation capacity and for capacity for catching up: A short review of the literature

What can we learn from the literature in terms of heterogeneity in innovative capacity and convergence in innovative capacity? (see also Veugelers & Schweiger (2016), Veugelers (2010) a.o.).

An important dimension explaining the heterogeneity across countries in their innovation capacity, is differences in levels of initial development and a country's initial position relative to the technology frontier (Verspagen (1991)¹²; Aghion & Howitt (1998)¹³).

For countries at still early stages of development, far from the technology frontier, the contribution of technology to growth runs through the country's ability to effectively absorb new technologies (World Bank (2008)¹⁴; Lall (2002)¹⁵). There are two key ingredients explaining differences across countries in effective technology take-up. The first is *access to foreign technology*. This requires openness through trade, foreign direct investment (FDI) and other forms of international cooperation. Second, access to (foreign) knowledge needs to be combined with a sufficiently developed 'absorptive capacity' (Cohen and Levinthal (1989)¹⁶) or 'social capability' (Abramovitz (1986)¹⁷), in order to deliver growth. This *absorptive capacity* depends on many factors, including the extent to which a country: has a technologically-literate workforce and a highly-skilled elite; promotes a pro-investment climate; permits the creation and expansion of firms using higher-technology processes; permits access to capital; and has adequate public-sector institutions to promote the take-up of critical technologies when private demand or market forces prove inadequate (World Bank (2008)).

For countries at higher levels of development, closer to the technology frontier, another factor that explains heterogeneity in innovation capacity comes into play, namely divergence in countries' *own indigenous innovative capacity* (Hoekman et al (2006)¹⁸). First, a country's own R&D complements the take-up of existing technology because it is a component of absorptive capacity. Foreign technologies frequently need modification so that they are suitable for domestic circumstances (Acemoğlu et al. (2006)¹⁹). But, at higher levels of development, a country's own R&D can increasingly substitute adoption of existing technologies, allowing the generation of new technologies, pushing the technology frontier. At this stage, countries require technological know-how, reflected in public and private R&D resources. They also need to be able to incentivise or reward innovation (e.g. Furman et al (2002)²⁰). In well-functioning *product markets*, that are

¹² Verspagen B. (1991) A new empirical approach to catching up or falling behind. *Structural Change and Economic Dynamics* 2(2): 359-380

¹³ Howitt P., & Aghion, P. (1998). Capital accumulation and innovation as complementary factors in long-run growth. *Journal of Economic Growth*, 3(2), 111-130

¹⁴ World Bank. (2008). Determinants of Technological Diffusion: Recent Trends and Prospects. *Global Economic Prospects 2008: Technology Diffusion in the Developing World*.

¹⁵ Lall S. (2002). FDI and development: research issues in the emerging context. *Foreign Direct Investment*, 339-359.

¹⁶ Cohen WM, Levinthal DA (1989) Innovation and Learning: The Two Faces of R&D. *Economic Journal* 99(397): 569-596

¹⁷ Abramovitz M (1986) Catching up, forging ahead and falling behind. *Journal of Economic History* 46(2): 386-406

¹⁸ Hoekman B, Smarzyska Javorcik B (eds.) (2006) *Global Integration and Technology Transfer*. New York, Palgrave MacMillan

¹⁹ Acemoğlu D, Aghion P, Zilibotti F (2006) Distance to frontier, selection, and economic growth. *Journal of the European Economic Association* 4(1): 37-74

²⁰ Furman JL, Porter ME, Stern S (2002) The determinants of national innovation capacity. *Research Policy* 31(6): 899-933

sufficiently open to enable competition between incumbent firms and the entry of new firms, incumbent firms will have incentives to innovate to improve their competitive position, while new firms can flow into the market, embodying new ideas. This also requires a large base of local or foreign customers willing to pay for innovative products, and effective *intellectual property rights* (IPR) schemes. Furthermore, new business opportunities can only be taken advantage of if appropriately educated and skilled workers can be hired under the right conditions. This requires the presence of skills and well-functioning *labour markets* that give innovators access to researchers and skilled human capital. Similarly, well-functioning *risk capital markets* give innovators access to capital to finance their innovative projects.

To summarise, a set of factors, shaping the country's capacity for a virtuous innovation growth link, need to be present. In addition to R&D, technology and IT infrastructure, these additional factors include access to large markets, (international) openness of markets, competition, access to a highly educated and skilled population (especially tertiary level), well-developed financial markets and, quality research institutions. It is important to note that these factors should not be seen in isolation, but as part of a system of key prerequisites for innovation capacity building.

The relative importance of each these factors in the required system depends on the level of initial development and a country's initial position relative to the technology frontier. We can therefore expect substantial heterogeneity among European countries, because of different capacities to put in place a virtuous innovation growth ecosystem, but also because of differences in initial conditions requiring differently composed ecoinnovation systems.

7. Indicators for assessing the potential of countries to catch up on innovation capacity

Following the literature (see sections above), the conditions influencing innovative capacity building can be divided into those affecting access to technology, those affecting firms' capacity to adopt and fully understand existing technologies, and those affecting the ability to create innovations new to markets. The importance of each of them will depend on differences in the position of a country on its path to development.²¹

Access to technology depends on a country's economic openness, the availability and use of ICT infrastructure and the extent to which openness (to FDI and trade) facilitates the transfer of technology.

Absorptive capacity is underpinned by the quality of secondary and undergraduate education and the effectiveness of on-the-job training.

What matters for creative capacity is: the quality of postgraduate education; the availability of highly qualified scientists and engineers; the quality of the public research infrastructure; effective cooperation between public research and industry; the protection of intellectual property and the availability of risk financing. Creative capacity is further underpinned by favorable incentives to engage in building innovative capacity. Business environment conditions for innovation include broader business environment conditions (or general framework conditions). Such conditions affect the operations and decisions of all firms in the economy, including firms that innovate. Business environment conditions include the quality of institutions and the functioning of product, labour and financial markets.

In what follows we will use evidence from the EU Innovation Scoreboard as indicators for the various dimensions composing a country's capacity for innovation capacity building. For a detailed description of the variables included, we refer to the EIS website.²²

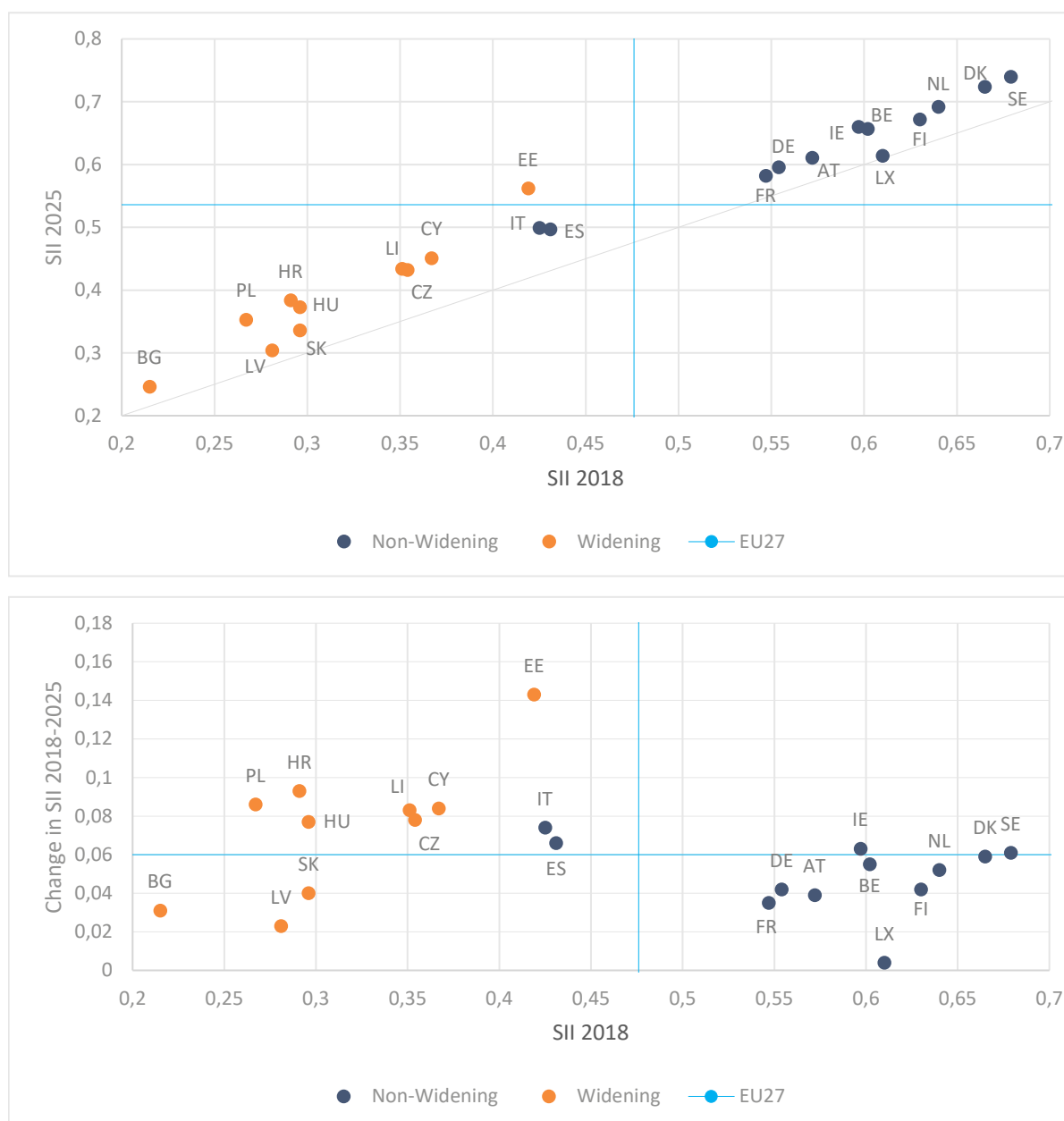
²¹ For a more detailed description and explanation of the various prerequisites and their measurement, see Veugelers (2011).

²² https://research-and-innovation.ec.europa.eu/statistics/performance-indicators/european-innovation-scoreboard_en

7.1. Assessing the position of EU 'Widening' countries on innovation capacity

The EIS evidence shows that Europe is not a homogeneous block when it comes to innovation. European countries are marked by heterogeneity in their innovation capacity. Although integration has resulted in some level of convergence in innovation, the pace of convergence is slow. There still remain substantial country differences in terms of capacity to build innovation capacity.

Figure 7 – The position and trend in EU Member States Innovation Capacity (SII 2018–2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data

The innovation divide in Europe does not follow a simple divide as can be seen on Figure 7. The weakest group of countries also includes some of the older Member States, most notably, Greece,

Spain, Italy and Portugal. At the same time, some of the transition countries, i.e. Estonia and Slovenia, have already made it into the lower-middle group of innovative countries. It also does not follow a simple development catching-up divide, with Italy as the clearest negative outlier as a developed European country with a low innovative capacity. Among the former cohesion countries, the catching up in innovative capacity differs substantially with Ireland as the best performer. Similar results were also found by Radosevic (2004)²³. In addition to a high-tech 'north' cluster composed of four countries with the highest national innovation capacities in EU (Finland, Sweden, Denmark and UK), he obtained two other clusters comprised of most of the catching-up Member States as well as some other Member States. One cluster is composed of the 3 cohesion states (Spain, Portugal and Greece) and 6 less advanced new Member States (Slovakia, Romania, Latvia, Lithuania, Poland and Bulgaria). They are characterised by rather weak national innovation capacities. The 4 more advanced new Member States (Czech Republic, Slovenia, Estonia and Hungary) together with 6 old Member States (Austria, Belgium, Germany, France, Italy and Ireland) form a kind of a 'middle level' group of the EU.

In what follows we will focus on the "Widening" dimension to sort Member States on their position in terms of innovation capacity. As already mentioned supra, all of the 15 'Widening' countries have for most of the time period considered (2018–2025) scored below the average EU27 SII score. Most of them have made some progress in reducing their gap with the EU27 average. This holds most notably for Estonia which has made most progress and scores above the EU27 average since 2021. Portugal was the best performing 'Widening' country in 2018, but has struggled to reduce its remaining gap. The countries at the bottom have made little progress and continue to be persistent laggards.

Table 4 – Evidence on convergence in innovation capacity within and between EU27 'Widening' countries

	2018	2025
CV SII EU27	0.32	0.27
Average SII EU27	0.44	0.50
Average SII 'Widening'	0.33	0.40
GAP in SII between 'Widening' and EU27 average	0.75	0.79
CV SII 'Widening'	0.25	0.25
TOP 3 SII 'Widening'	PT; MT; SI	EE; MT; SI
BOTTOM 3 SII 'Widening'	PL; BU; RO	LV; BU; RO

Source: Own calculations based on [EC-JRC EIS2025](#) data.

Overall, 'Widening' countries have managed to reduce their gap in their SII score over time within the EU27 ('beta-convergence', see Figure 2b), but only slightly so. Similarly, the coefficient of variation in SII scores within the EU27 has somewhat decreased over time, illustrating sigma-convergence, but also only slightly. The variation within the group of 'Widening' countries is also substantial and persistent over time.

²³ Radosevic, S. (2004). A two-tier or multi-tier Europe? Assessing the innovation capacities of Central and East European countries in the enlarged EU. *JCMS: Journal of Common Market Studies*, 42(3), 641–666.

7.2. Assessing the potential of EU 'Widening' countries to catch up on innovation capacity

We will now zoom in on the various subcomponents of the European Innovation Scoreboard as relevant for identifying countries on their capacity to catch up, by assessing how 'Widening' countries score on factors important for shaping innovation capacity building.

7.2.1. ABSORPTIVE CAPACITY (ABS CAP)

To measure absorptive capacity, we use the following European Innovation Scoreboard factors:

- 1.1.2 Population aged 25–34 with tertiary education
- 1.1.3 Lifelong learning
- 1.3.2 Individuals who have above basic overall digital skills
- 2.3.2 Employed ICT specialists

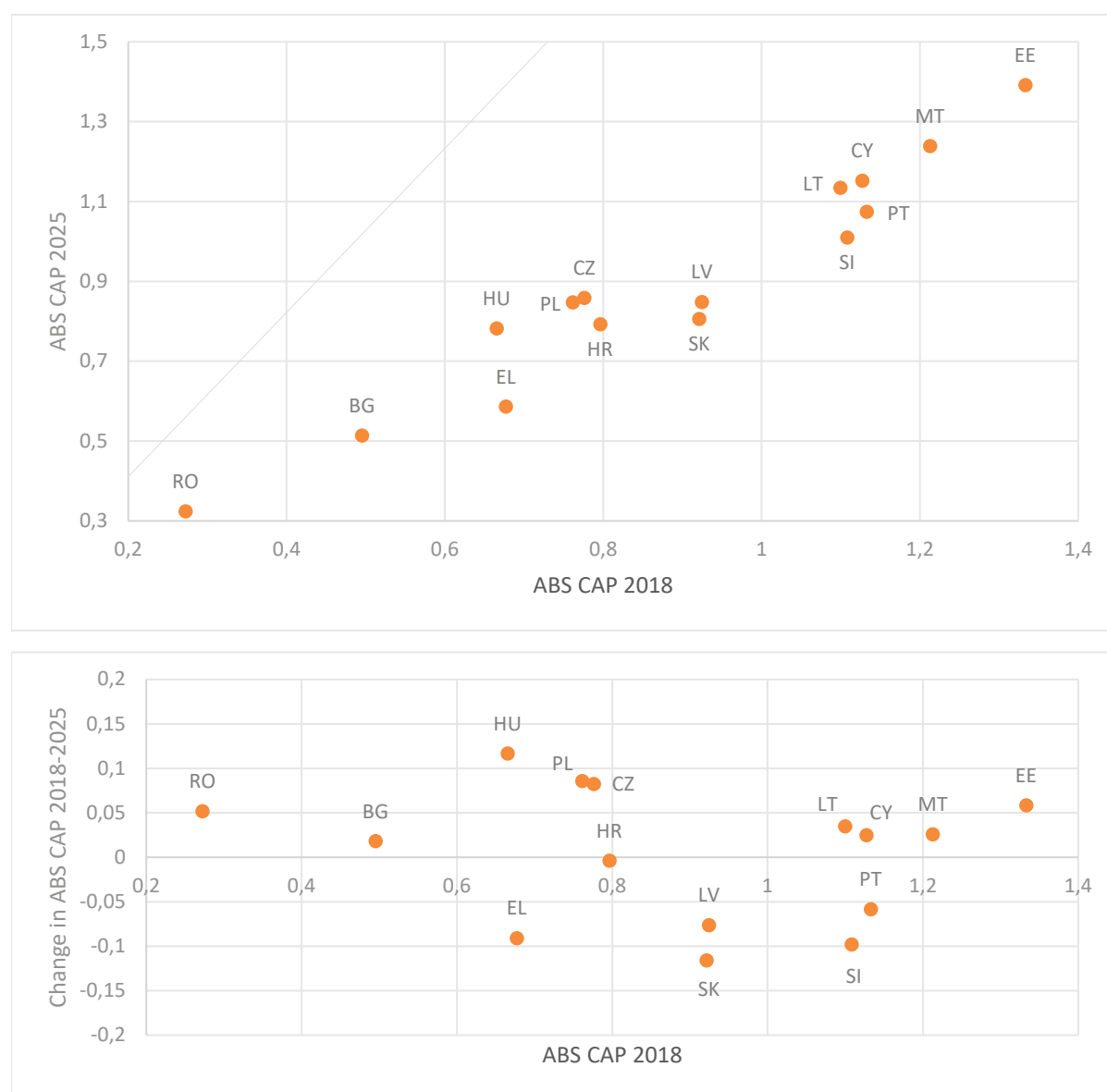
We take the absorptive capacity average score of countries along each of these 4 factors, weighing each of the 4 factors as equally important.

Table 5 – Evidence on position and convergence in absorptive capacity within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018					2025				
	1.1.2	1.1.3	1.3.2	2.3.2	ABS CAP	1.1.2	1.1.3	1.3.2	2.3.2	ABS CAP
CV EU27	0,62	0,72	0,45	0,40	0,44	0,54	0,57	0,47	0,34	0,38
GAP between 'Widening' and EU27 average	0,86	0,69	0,77	0,80	0,78	0,84	0,79	0,80	0,86	0,82
CV 'Widening'	0,53	0,61	0,39	0,31	0,33	0,54	0,53	0,45	0,30	0,32
GAP between Bottom3 and Top3 'Widening'	0,40	0,26	0,33	0,55	0,39	0,60	0,21	0,31	0,49	0,39

Source: Own calculations based on [EC-JRC EIS2025](#) data.

Figure 8 – 'Widening' Countries gap with EU27 average on Absorptive Capacity (2018-2025)



Our findings on Absorptive Capacity are the following:

- The gap of the 'Widening' countries in Absorptive Capacity is somewhat smaller than the overall SII gap.
- There is catching up (beta-convergence), but it is slow.
- There is a higher variance in Absorptive Capacity than overall for SII, both in the EU27 and among the 'Widening' countries.
- There is sigma convergence in the EU, but not within 'Widening' countries.
- The TOP3 in Absorptive Capacity among 'Widening' countries are also the TOP3 in overall SII (EE, MT, CY). Romania and Bulgaria are stable bottom scorers. EL, SK, SI saw their gap increasing over time.

7.2.2. Science & Technology (S&T) ACCESS

To measure the access to science & technology, we use the following EIS factors:

- 1.2.1 International scientific co-publications
- 1.2.3 Foreign doctorate students
- 1.3.1 High Speed Internet Access
- 2.3.1 Cloud computing in enterprises (not available for 2018)
- 2.3.2 Employed ICT specialists
- 4.2.3 High tech imports from partners outside of the EU

We take the average science & technology score of countries along each of these 5 factors, weighing each of the 5 factors as equally important.

Table 6 – Evidence on position and convergence in S&T Access within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018						2025						
	1.2.1	1.2.3	1.3.1	2.3.2	4.2.3	S&T ACC	1.2.1	1.2.3	1.3.1	2.3.1	2.3.2	4.2.3	S&T ACC
CV EU27	0,78	0,90	0,79	0,40	0,30	0,35	0,75	0,76	0,26	0,40	0,34	0,32	0,26
GAP between 'Widening' and EU27 average	0,65	0,47	0,92	0,80	0,86	0,77	0,74	0,76	0,94	0,87	0,86	0,89	0,85
CV 'Widening'	0,57	0,71	0,63	0,31	0,41	0,25	0,60	0,83	0,29	0,41	0,30	0,56	0,25
GAP between Bottom3 and Top3 'Widening'	0,81	0,55	0,12	0,63	0,72	0,51	0,41	0,09	0,74	0,40	0,52	0,96	0,51

Source: Own calculations based on [EC-JRC EIS2025](#) data.

Another important channel for S&T access noted in the literature (cf supra). The SII index does not include FDI or share of foreign firms in GDP. Using EUROSTAT data on the presence of foreign firms in the 'Widening' countries shows that this is a substantial factor in HU, CZ, SK, which would improve these countries position in S&T access, if including this factor (see Figure 10 below). It is also important for the bottom scoring countries BG, RO, LV, raising their scoring on S&T access. It is less important for CY and EL.

Figure 9 – 'Widening' Countries gap with EU27 average on S&T Access (2018–2025)

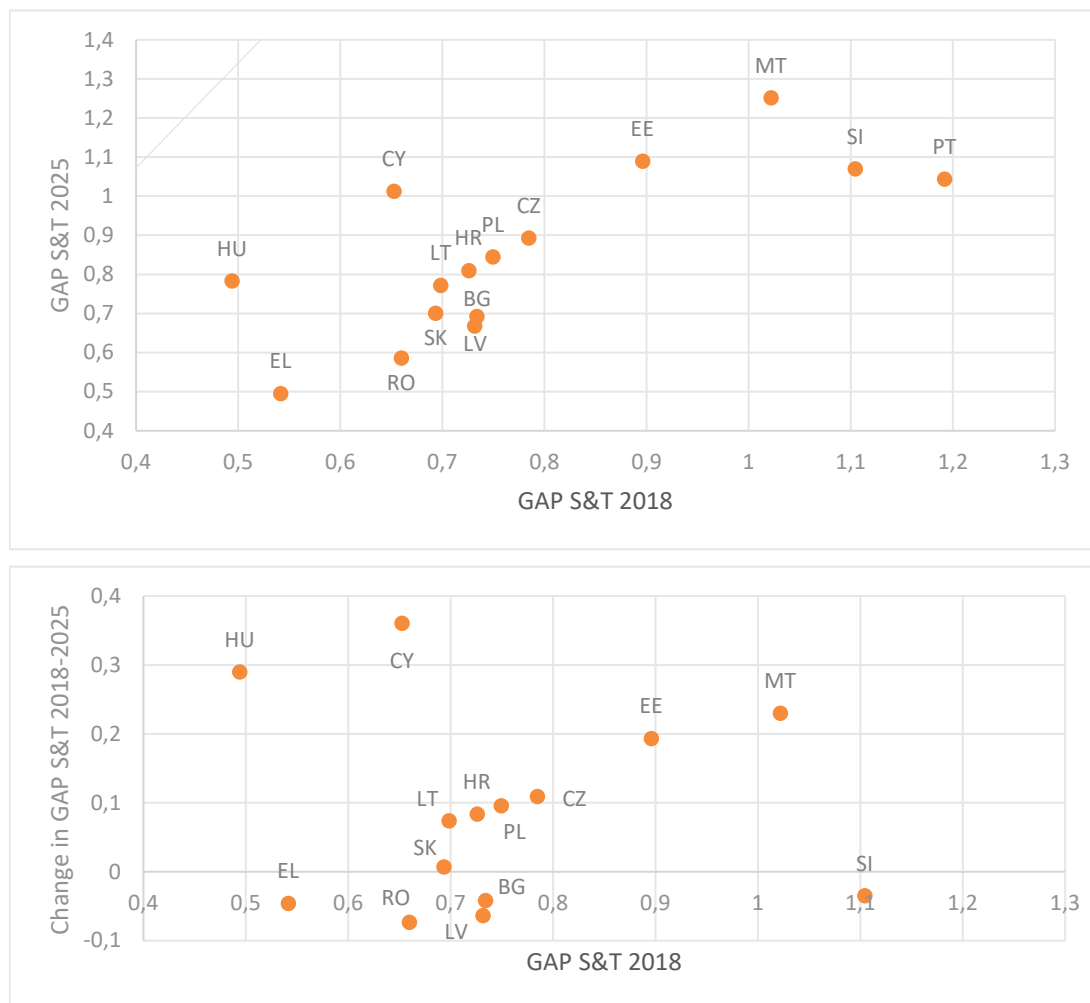
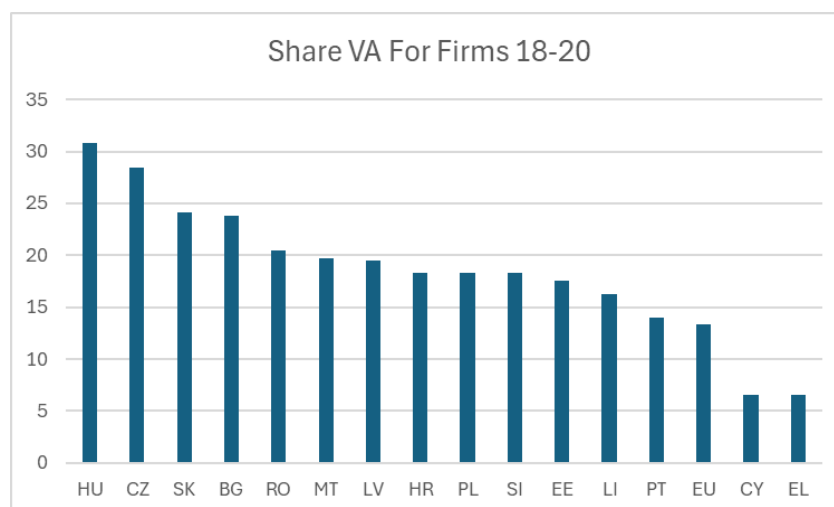


Figure 10 – 'Widening' Countries Foreign Firms presence

Source: [EUROSTAT](https://ec.europa.eu/eurostat)

Our findings on S&T Access are the following:

- The gap of the 'Widening' countries in S&T Access is not much different from the overall SII gap.
- The highest gap and variance can be found for 1.2.1 and 1.2.3 (Science access).
- The gap of the 'Widening' countries in S&T Access has become smaller. There is some catching up (beta-convergence).
- There is sigma convergence in the EU, but not among 'Widening' countries.
- The gap between the TOP3 in S&T Access and the BOTTOM3 has not changed much over time. The top SII countries EE and SI also score top on S&T Access. EL joins the BOTTOM3 on S&T Access.

7.2.3. QUALITY OF THE RESEARCH SYSTEM (RES SYS)

To measure the quality of the research system, we use the following EIS factors:

- 1.1.1 New doctorate graduates (in STEM)
- 1.2.1 International scientific co-publications
- 1.2.2 Top 10% most cited publications
- 1.2.3 Foreign doctoral students

We take the average score of countries along each of these 4 factors, weighing each of the 4 factors as equally important.

For 2025 we also look at the quality of the Higher Education Institutes of a country by looking ranking of Higher Education Institutes from Times Higher Education (THE). We produce a country's score depending on the position of the HEIs of the country in the THE ranking 2025 (See Box for further explanations on this indicator).

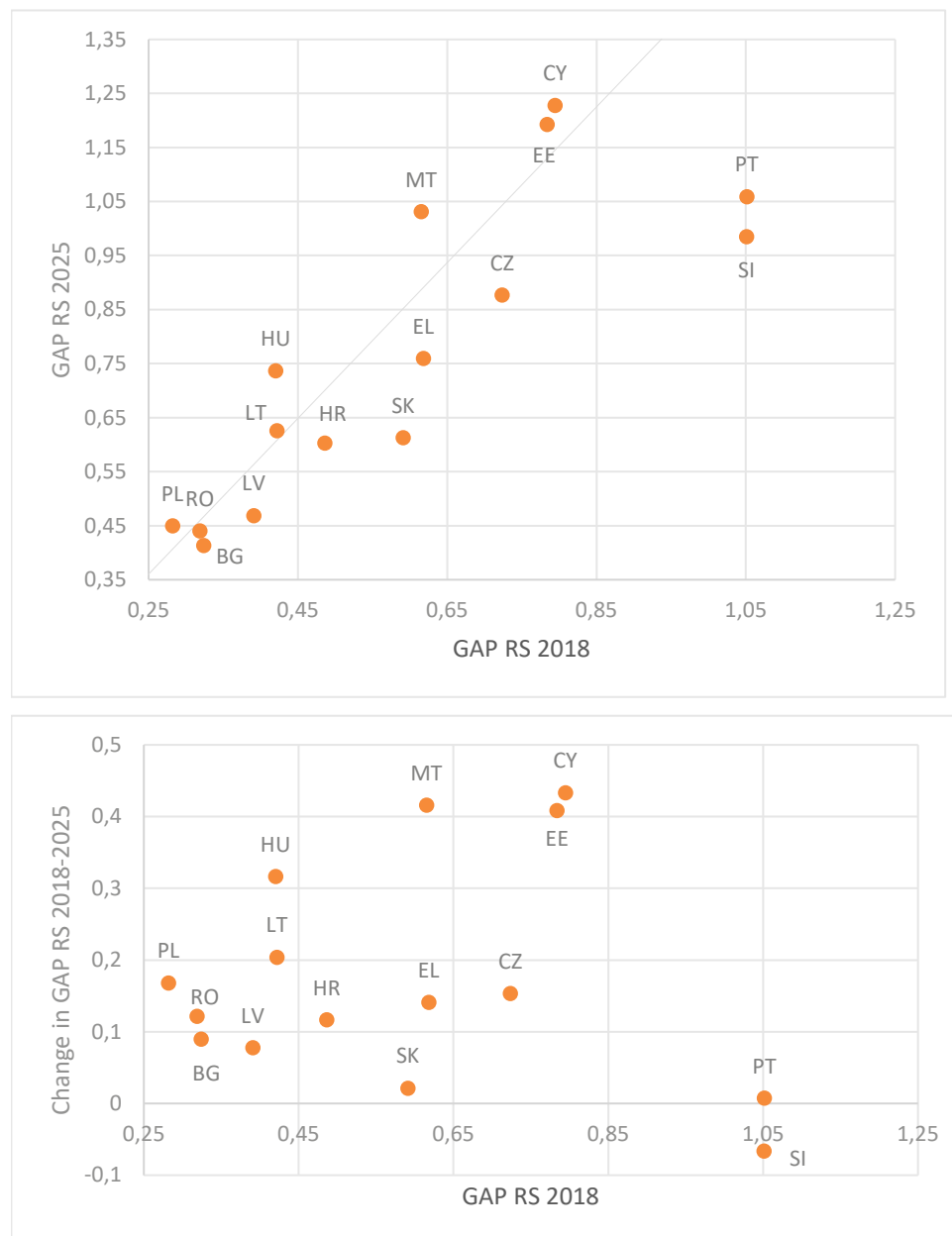
Table 7 – Evidence on position and convergence in Quality of the Research System within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018					2025					
	1.1.1	1.2.1	1.2.2	1.2.3	RES SYS	1.1.1	1.2.1	1.2.2	1.2.3	RES SYS	THE
CV EU27	0,44	0,78	0,45	0,90	0,49	0,41	0,75	0,38	0,76	0,45	0,28
GAP between 'Widening' and EU27 average	0,72	0,65	0,61	0,47	0,62	0,71	0,74	0,71	0,76	0,73	0,80
CV 'Widening'	0,59	0,57	0,59	0,71	0,42	0,46	0,60	0,41	0,83	0,42	0,17
GAP between Bottom3 and Top3 'Widening'	0,52	0,18	0,29	0,18	0,32	0,50	0,19	0,47	0,17	0,32	0,68

Source: Own calculations based on [EC-JRC EIS2025](#) data.

For the quality of the higher education institutes (Figure 11 below), as measured by their THE ranking for 2025, we get a very similar ranking of 'Widening' countries as for our aggregate RES SYS, with CY, EE and PT in the TOP and RO & BG in the BOTTOM. SK's gap on THE is somewhat larger than its gap on RES SYS.

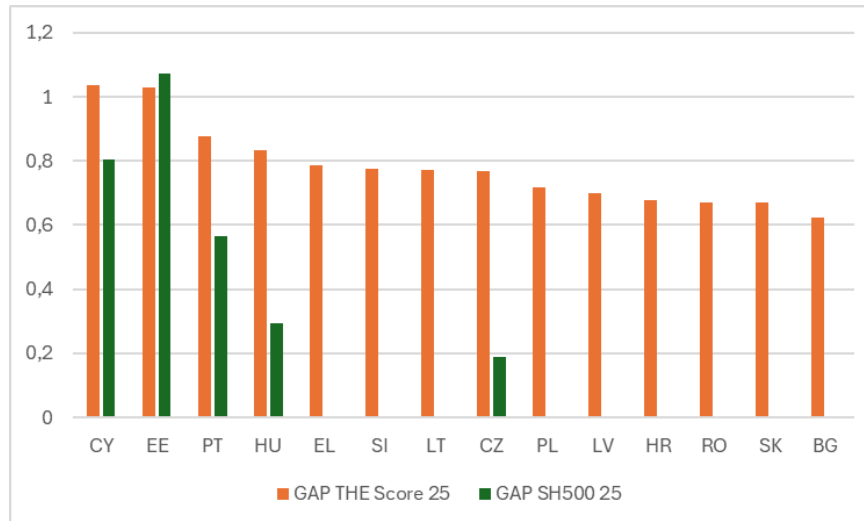
Figure 11 – 'Widening' Countries gap with EU27 average on Quality of the Research System (2018-2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

We also look at the share of HEI that are in the THE TOP500, for which only the TOP 'Widening' countries have a positive score. EE scores particularly well on this, even slightly above the EU27 average.

Figure 12 – 'Widening' Countries gap with EU27 average on Quality of Higher Education 2025



NB: MT not included as having only 1 institution in THE ranking

Source: Own calculations based on [Times Higher Education](#), Edition 2025

Our findings on Quality of the Research System are the following:

- The gap of the 'Widening' countries in RES SYS is substantial and substantially higher than their overall SII gap.
- The gap of the 'Widening' countries in RES SYS has become smaller. There is beta-convergence.
- There is a higher variance in RES SYS than overall for SII, both in the EU27 and among the 'Widening' countries. The THE score has the lowest variance.
- There is a slight sigma convergence in the EU.
- There is no sigma converge among the 'Widening' countries.
- The gap between the TOP3 in RES SYS and the BOTTOM3 has not changed over time. EE, CY and PT are TOP 'Widening' countries on this indicator; EE, CY, MT and HU have improved their gap score the most, while RO, BG, LV are the bottom scorers, joined by PL.

7.2.4. INNOVATION ACTIVE BUSINESS SECTOR (INN ENT)

To measure how innovative active enterprises are, we use the following EIS factors:

2.2.1 R&D expenditures in the business sector

2.2.3 Innovative expenditures per person employed in innovation active enterprises

4.1.1 Sales of product innovations

We also include

ENT Share of enterprises that are innovation active (Source: EC-JRC IOI database²⁴)

We take the average score of countries (innovation active business sector) along each of these 4 factors, weighing each of the 4 factors as equally important.

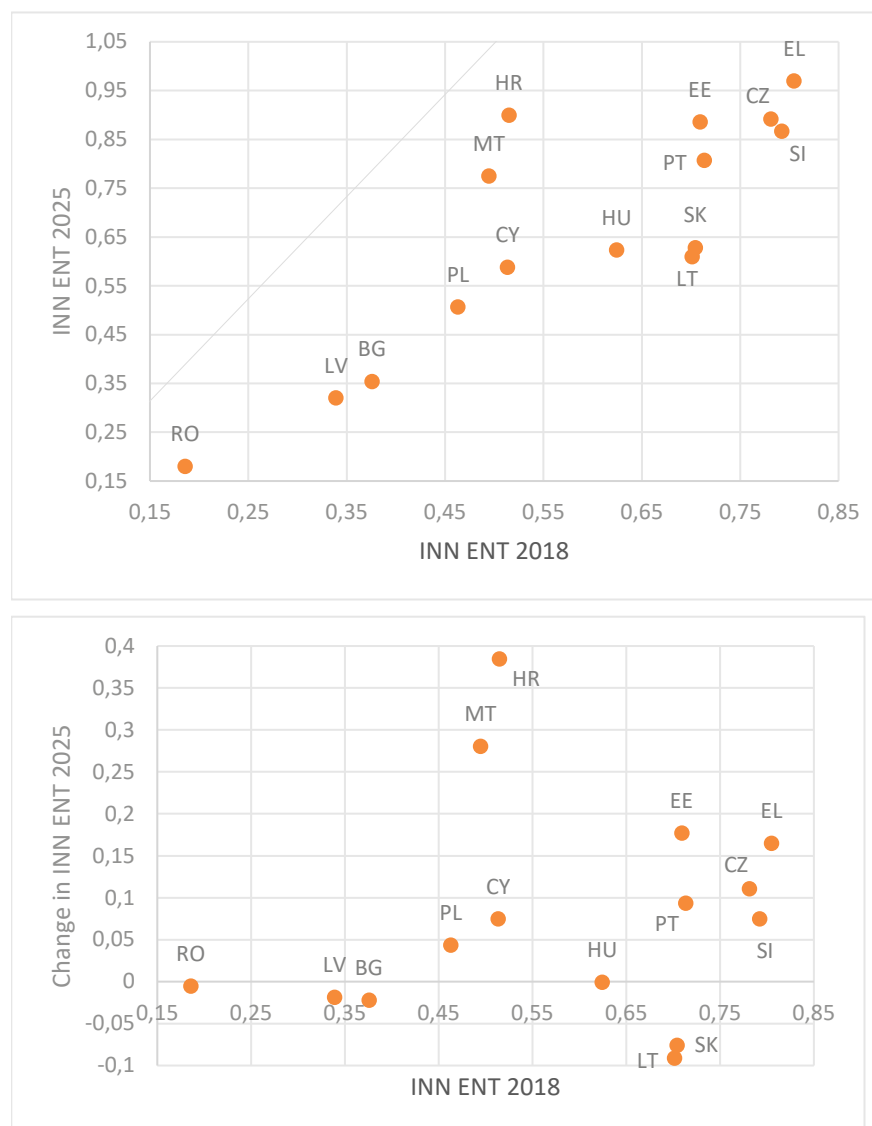
Table 8 – Evidence on position and convergence in Innovative Enterprises within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018					2025				
	2.2.1	2.2.3	4.1.1	ENT	INN ENT	2.2.1	2.2.3	4.1.1	ENT	INN ENT
CV EU27	0,51	0,49	0,38	0,31	0,34	0,47	0,47	0,45	0,28	0,33
GAP between 'Widening' and EU27 average	0,55	0,57	0,91	0,80	0,71	0,64	0,69	0,88	0,90	0,78
CV 'Widening'	0,74	0,46	0,45	0,35	0,32	0,54	0,70	0,48	0,35	0,36
GAP between Bottom3 and Top3 'Widening'	0,29	0,23	0,46	0,47	0,38	0,39	0,16	0,27	0,43	0,31

Source: Own calculations based on [EC-JRC EIS2025](#) data.

²⁴ [Regional Innovation Output Indicator, DG_JRC](#)

Figure 13 – 'Widening' Countries gap with EU27 average on INN ENT (2018–2025)



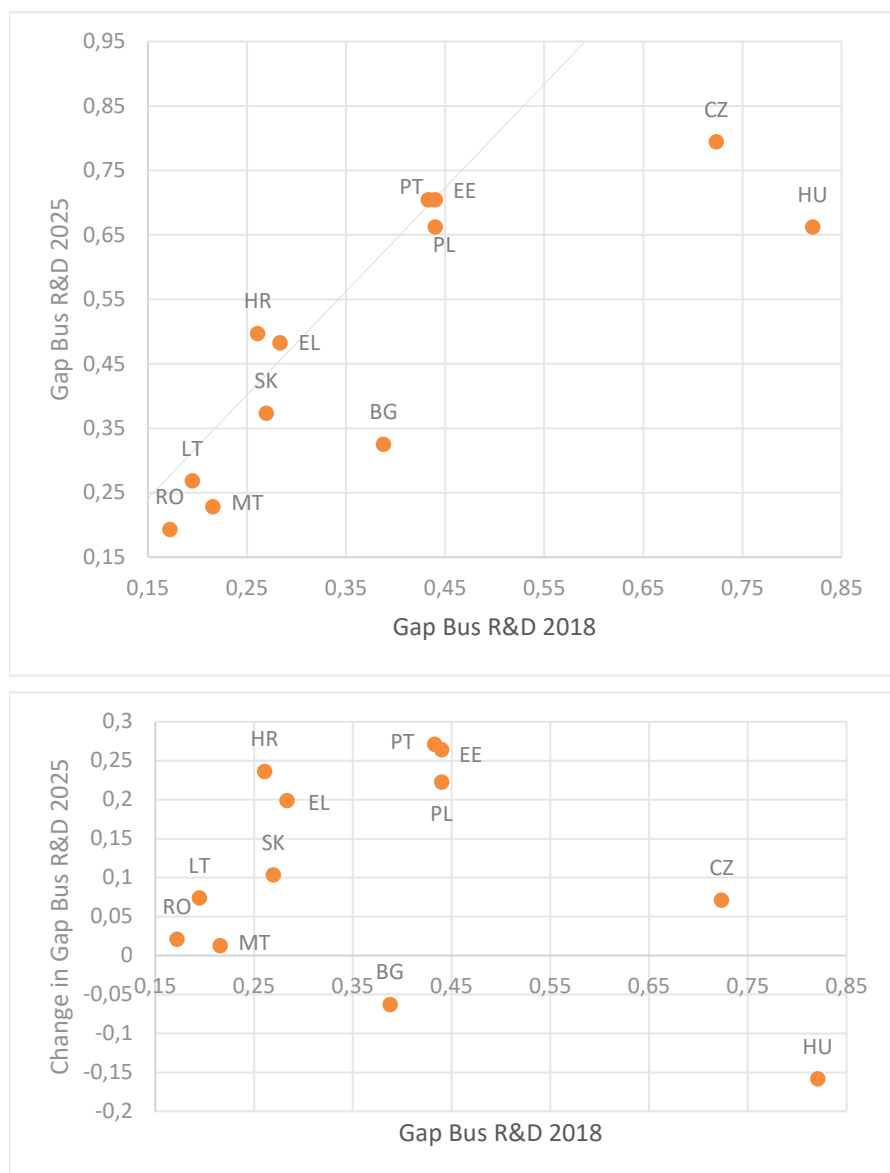
Source: Own calculations based on [EC-JRC EIS2025](#) data.

Our findings on Innovation Active Enterprises are the following:

- The gap of the 'Widening' countries in INN ENT is somewhat smaller than the overall SII gap.
- The gap of the 'Widening' countries in INN ENT has become quite substantially smaller. There is quite some catching up (beta-convergence).
- There is a higher variance in INN ENT than overall for SII, both in the EU27 and among the 'Widening' countries.
- There is sigma convergence in the EU, but not within 'Widening' countries.
- The gap between the TOP3 in INN ENT and the BOTTOM3 has increased over time. HR, MT, but also EE and EL have improved their score, while RO, BG, LV are stable bottom scorers.

R&D spent by the business sector (factor 2.2.1) is a highly used indicator to reflect on EU's innovation deficit. We therefore zoom in on this indicator separately.

Figure 14 – 'Widening' Countries gap with EU27 average on Business R&D (2018-2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

On this indicator SI, CZ and EE are the top performing 'Widening' countries, with especially Estonia's catching up on this indicator noteworthy. CY and EL loose substantial positions on this factor when only focusing on this indicator.

7.2.5. LINKAGES (LINK)

To measure linkages, we include the following EIS factors:

- 1.2.1. International scientific co-publications
- 1.2.3. Foreign doctoral students
- 3.2.1. Innovative SMEs collaborating with others
- 3.2.2. Public private co-publications
- 3.2.3 Job-to-job mobility of Human Resources in Science

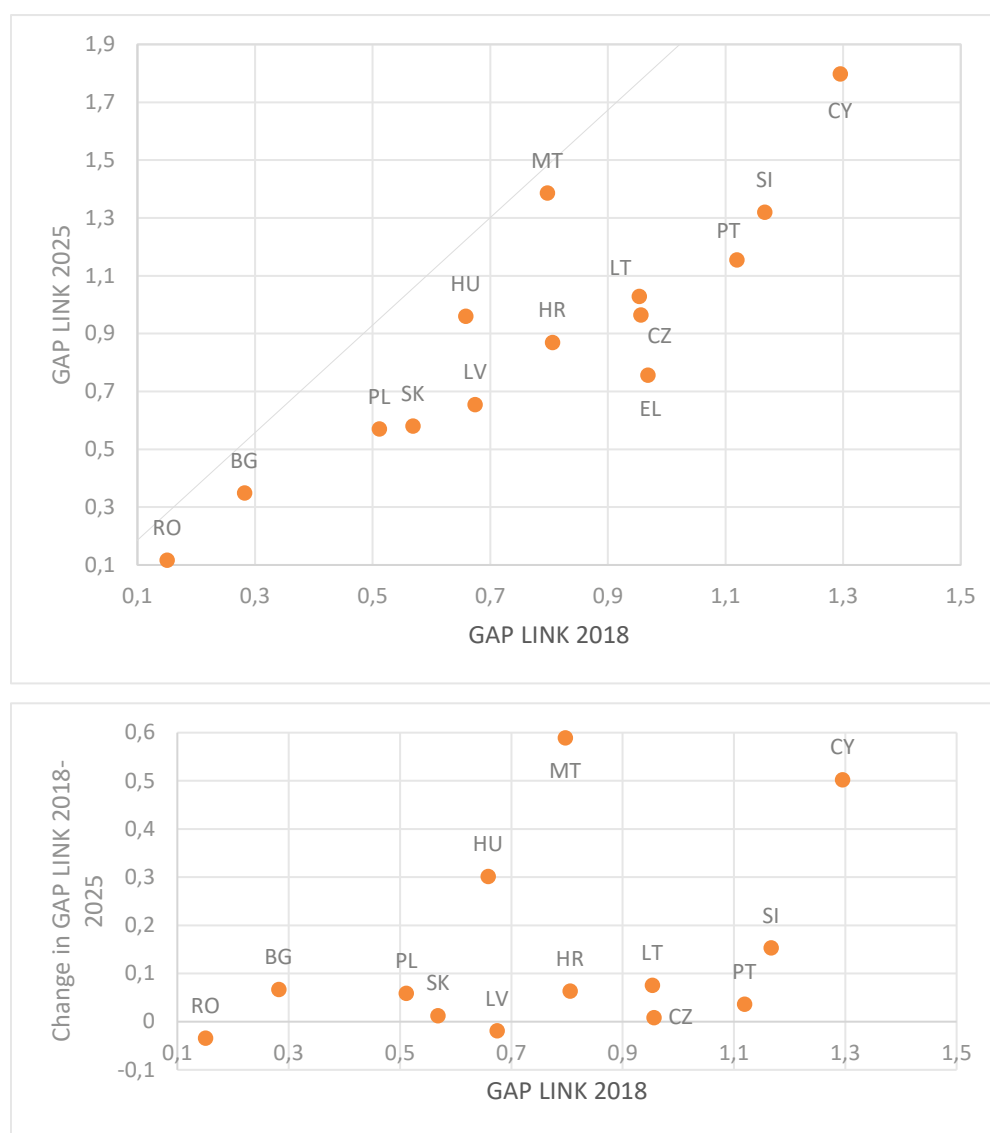
We take the average score of countries along each of these 5 factors, weighing each of the 5 factors as equally important.

Table 9 – Evidence on position and convergence in Linkages within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018						2025					
	1.2.1	1.2.3	3.2.1	3.2.2	3.2.3	LINK	1.2.1	1.2.3	3.2.1	3.2.2	3.2.3	LINK
CV EU27	0,78	0,90	0,79	1,29	0,66	0,69	0,75	0,76	0,59	1,26	0,50	0,57
GAP between 'Widening' and EU27 average	0,65	0,47	0,85	0,56	0,71	0,66	0,74	0,76	0,71	0,66	0,91	0,76
CV 'Widening'	0,57	0,71	0,77	0,56	0,57	0,46	0,60	0,83	0,50	0,56	0,59	0,49
GAP between Bottom3 and Top3 'Widening'	0,17	0,25	0,15	0,18	0,40	0,23	0,20	0,13	0,30	0,21	0,26	0,21

Source: Own calculations based on [EC-JRC EIS2025](#) data.

Figure 15 – 'Widening' Countries gap with EU27 average on LINK (2018–2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data

Our findings on LINKAGES are the following:

- The gap of the 'Widening' countries in LINK is larger than the overall SII gap.
- The gap of the 'Widening' countries in LINK has become quite substantially smaller. There is quite some catching up (beta-convergence);
- There is a much higher variance in LINK than overall for SII. In the EU27 this holds especially for public private co-publications; Among the 'Widening' countries this holds especially true for foreign doct students and innovative SMEs collaborating.
- There is sigma convergence in the EU, but divergence among 'Widening' countries.
- The gap in LINK between the TOP3 and the BOTTOM3 among 'Widening' countries is substantial and has somewhat increased over time. CY, MT have made substantial progress, EE is a stable Top scorer among 'Widening' countries; The BOTTOM are stable bottom scorers, with RO having a very substantial and stable gap.

7.2.6. INTELLECTUAL PROPERTY (IP)

To measure progress in intellectual property domain, we include the following EIS factors:

3.3.1 PCT patent applications

3.3.2 Trademark applications

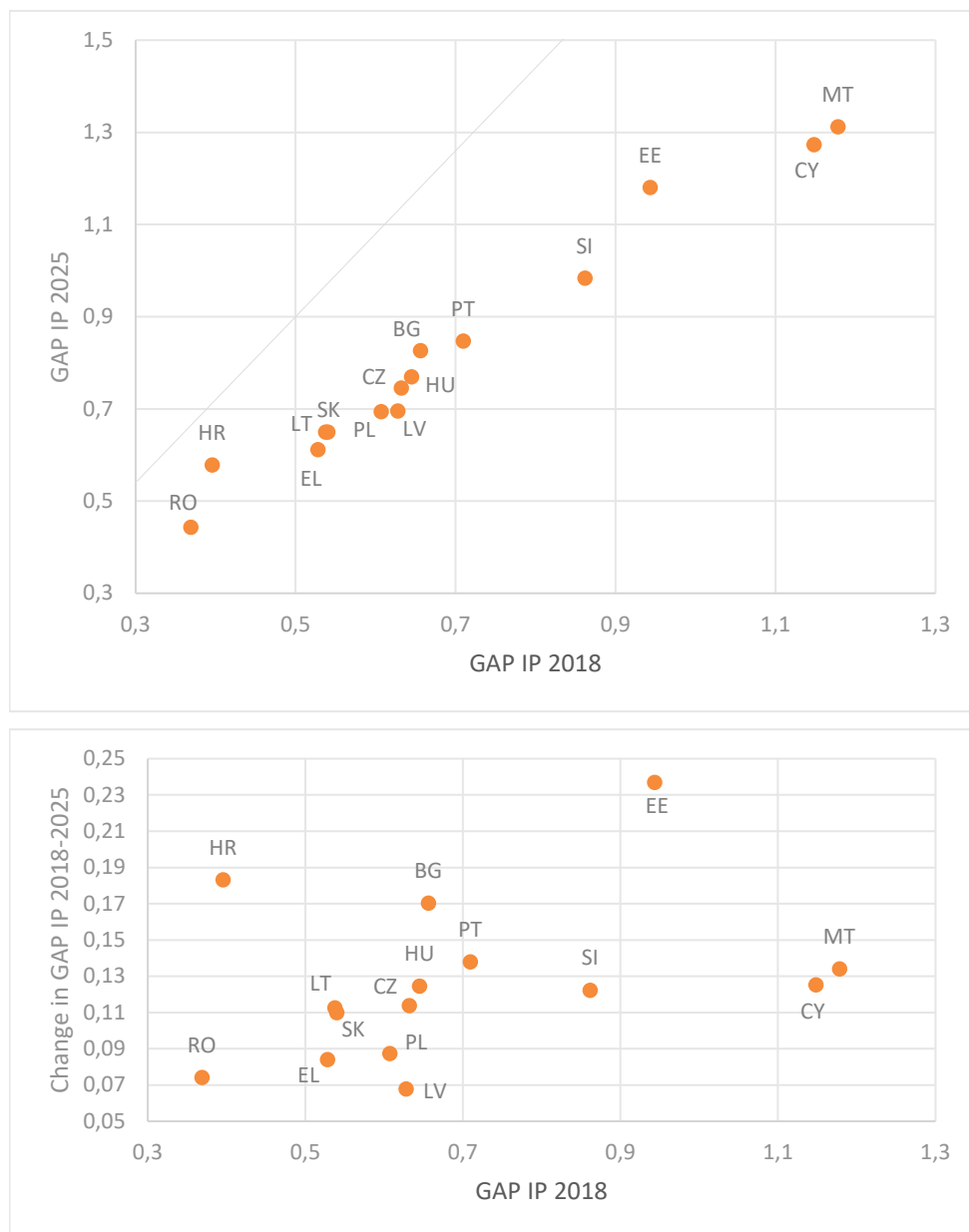
We take the average score of countries along each of these 2 factors, weighing each of the 2 factors as equally important.

Table 9 – Evidence on position and convergence in IP within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018			2025		
	3.3.1	3.3.2	IP	3.3.1	3.3.2	IP
CV EU27	0,37	0,48	0,31	0,40	0,46	0,29
GAP between 'Widening' and EU27 average	0,64	0,94	0,78	0,65	1,07	0,86
CV 'Widening'	0,24	0,55	0,35	0,27	0,45	0,32
GAP between Bottom3 and Top3 'Widening'	0,68	0,28	0,40	0,69	0,34	0,43

Source: Own calculations based on [EC-JRC EIS2025](#) data.

Figure 16 – 'Widening' Countries gap with EU27 average on IP (2018-2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

Our findings on IP are the following:

- The gap of the 'Widening' countries in IP is somewhat smaller than the overall SII gap.
- The gap of the 'Widening' countries in IP has become smaller. There is catching up (beta-convergence).
- There is a higher variance in IP than overall for SII for the 'Widening' countries.
- There is slight sigma convergence in the EU27 and in 'Widening' countries.
- With all 'Widening' countries improving their relative score, the gap between the TOP3 and the BOTTOM3 becomes somewhat smaller over time. Notably, BG is for IP not in the BOTTOM3.

7.2.7. ACCESS TO FINANCE (FIN)

To measure access to finance for innovation, we include the following EIS factors:

2.1.1 R&D expenditures in the public sector

2.1.2 Venture capital expenditures

2.1.3 Direct government funding and government tax support for business R&D

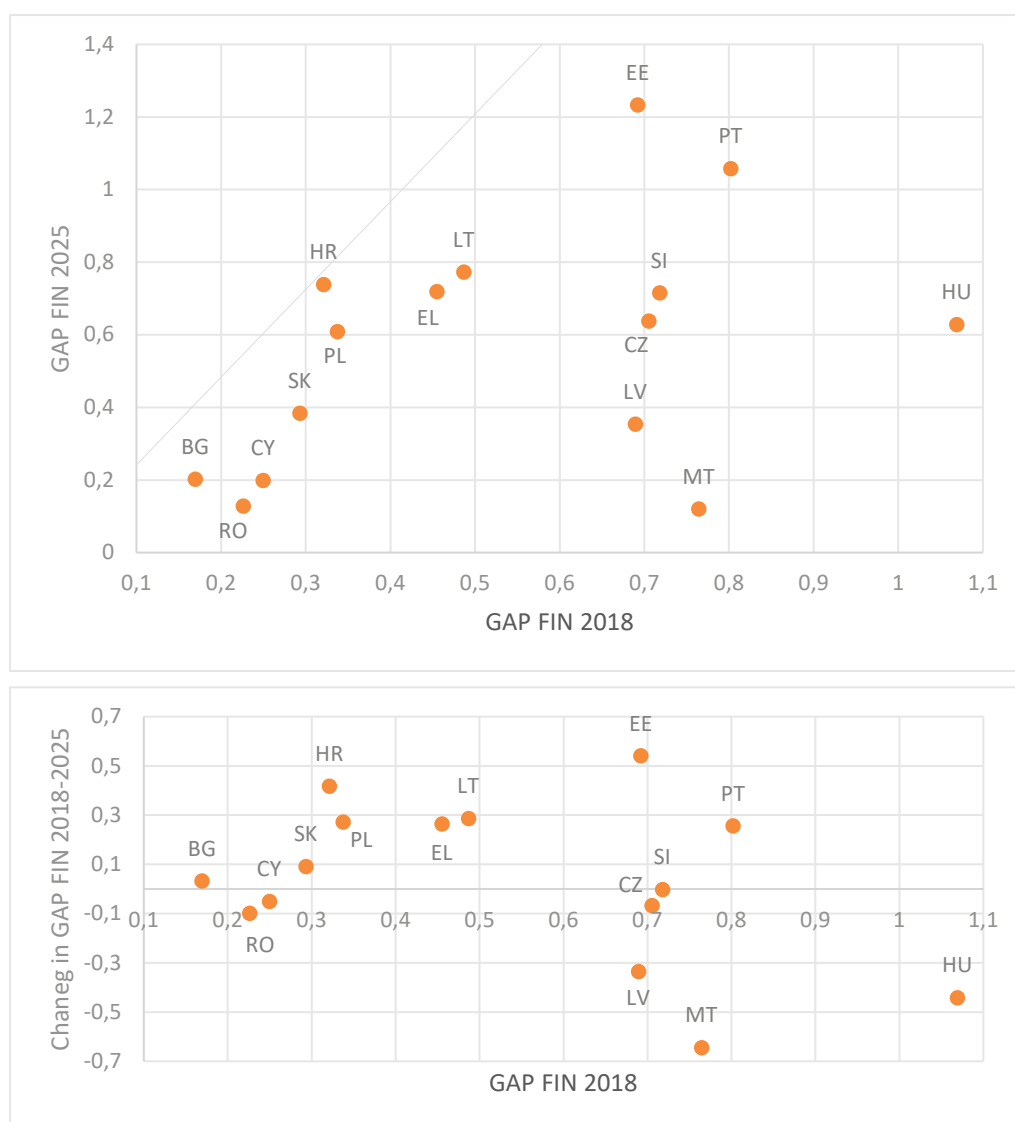
We take the average score of countries along each of these 3 factors, weighing each of the 3 factors as equally important.

Table 10 – Evidence on position and convergence in Access to Finance within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018				2025			
	2.1.1	2.1.2	2.1.3	FIN	2.1.1	2.1.2	2.1.3	FIN
CV EU27	0,42	0,87	0,65	0,38	0,41	0,81	0,58	0,42
GAP between 'Widening' and EU27 average	0,67	0,78	0,63	0,69	0,77	0,62	0,67	0,70
CV 'Widening'	0,53	1,23	1,29	0,50	0,47	1,24	1,06	0,59
GAP between Bottom3 and Top3 'Widening'	0,39	0,32	0,10	0,24	0,25	0,07	0,11	0,15

Source: Own calculations based on [EC-JRC EIS2025](#) data.

Figure 17 – 'Widening' Countries gap with EU27 average on FIN (2018-2025)



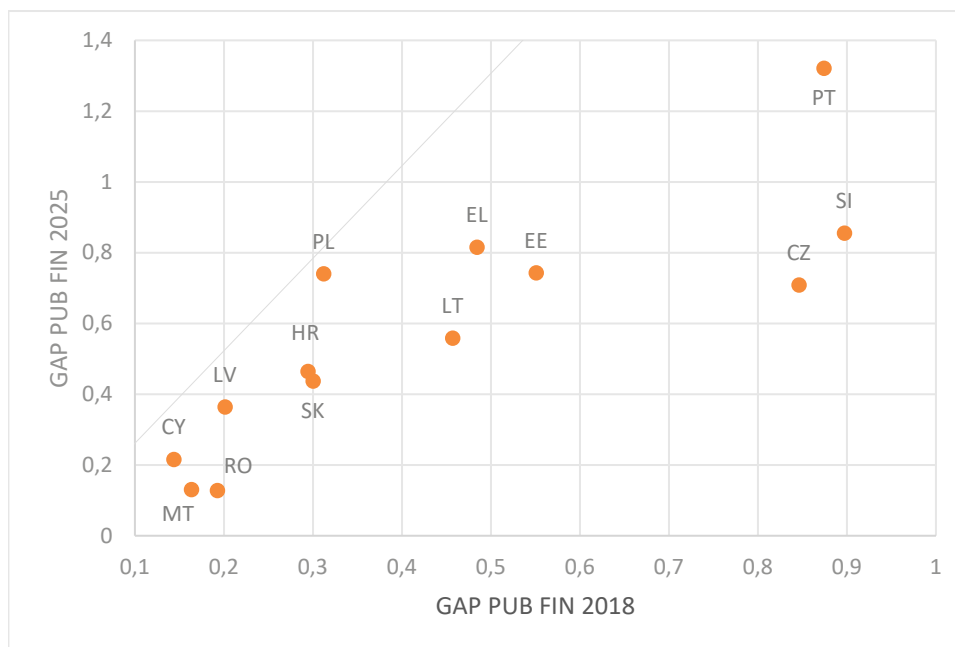
Source: Own calculations based on [EC-JRC EIS2025](#) data.

Our findings on Access to Finance for Innovation are the following:

- The gap of the 'Widening' countries in FIN is substantially bigger than their overall SII gap.
- The gap is especially big in 2.1.3 and more recently also in 2.1.2.
- There is no catching up (no beta-convergence).
- There is a substantially higher variance in FIN than overall for SII within the EU overall and also within the 'Widening' countries.
- There is no sigma convergence, even a sigma divergence in the EU27 and in 'Widening' countries.
- The gap in FIN between the TOP3 and the BOTTOM3 among 'Widening' countries is substantial and has increased, especially with the TOP improving their score and the BOTTOM staying put or deteriorating (eg MT).

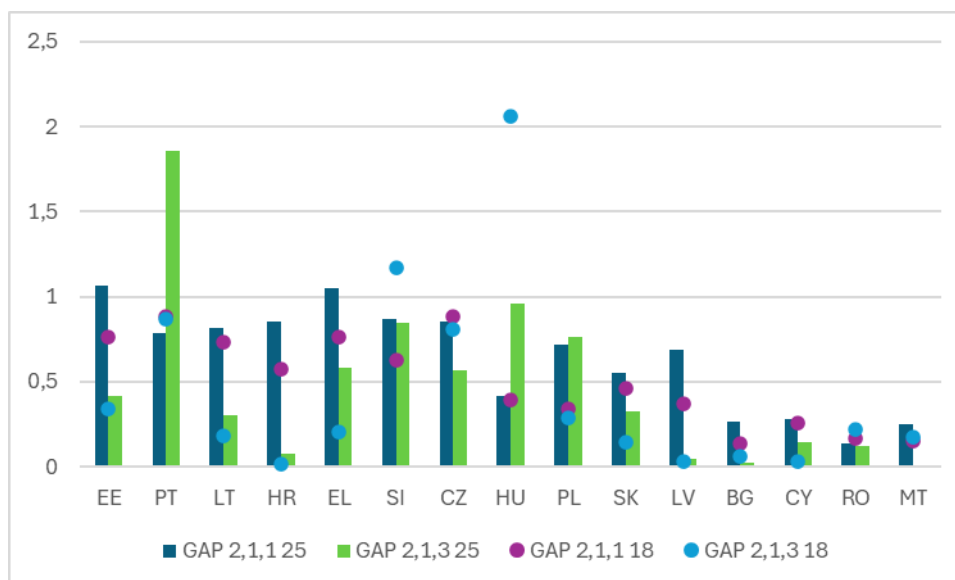
In the following, we zoom in only the public financing for innovation, ie EIS factors 2.1.1. en 2.1.3: jointly in PUB FIN; and separately.

Figure 18 – 'Widening' Countries gap with EU27 average on Public Finance for R&D (2018–2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

Figure 19 – 'Widening' Countries gap with EU27 average on 2.1.1 and 2.1.3 (2018–2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

7.2.8. IMPACT (EFF)

To assess the impact of innovation, we look at the following EIS factors:

- 4.2.1 Medium and High tech product exports
- 4.2.2 Knowledge intensive services exports
- 4.2.3 High tech imports from partners outside of the EU
- 4.3.3 Labour productivity (PROD)

We take the average score of countries along each of these 4 factors, weighing each of the 4 factors as equally important.

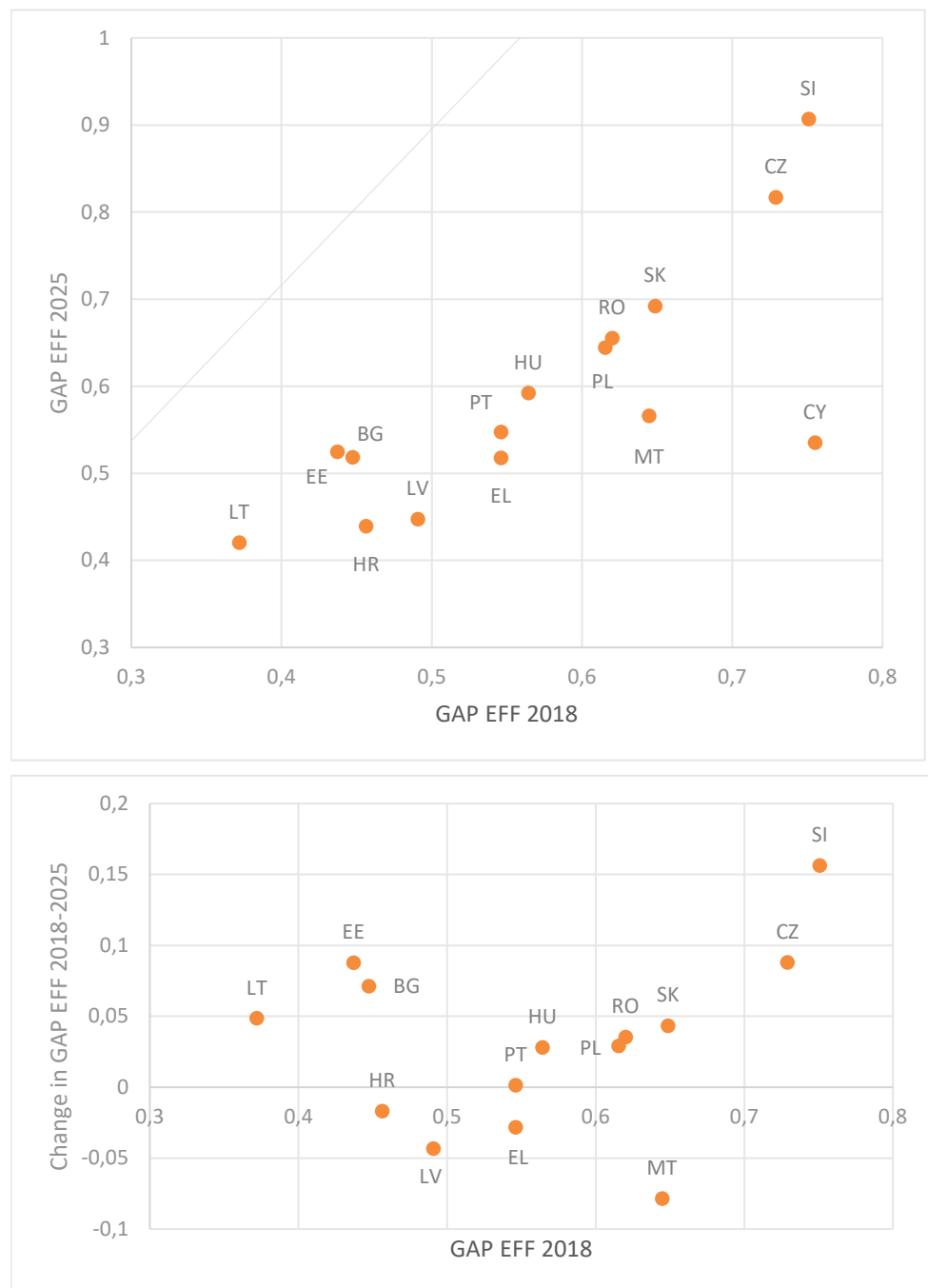
Table 11 – Evidence on position and convergence in Impact within and between 'Widening' countries and EU27 detailing Coefficient of Variation (CV) of EIS factors and the gaps between countries subsets.

	2018					2025				
	4.2.1	4.2.2	4.2.3	PRO D	EFF	4.2.1	4.2.2	4.2.3	PRO D	EFF
CV EU27	0,25	0,34	0,30	0,71	0,23	0,27	0,32	0,32	0,70	0,22
GAP between 'Widening' and EU27 average	0,94	0,72	0,86	0,40	0,77	0,94	0,78	0,89	0,43	0,80
CV 'Widening'	0,44	0,46	0,41	0,48	0,21	0,47	0,41	0,56	0,43	0,23
GAP between Bottom3 and Top3 'Widening'	0,48	0,63	0,66	0,43	0,56	0,38	0,83	0,60	0,58	0,54

Source: Own calculations based on [EC-JRC EIS2025](#) data.

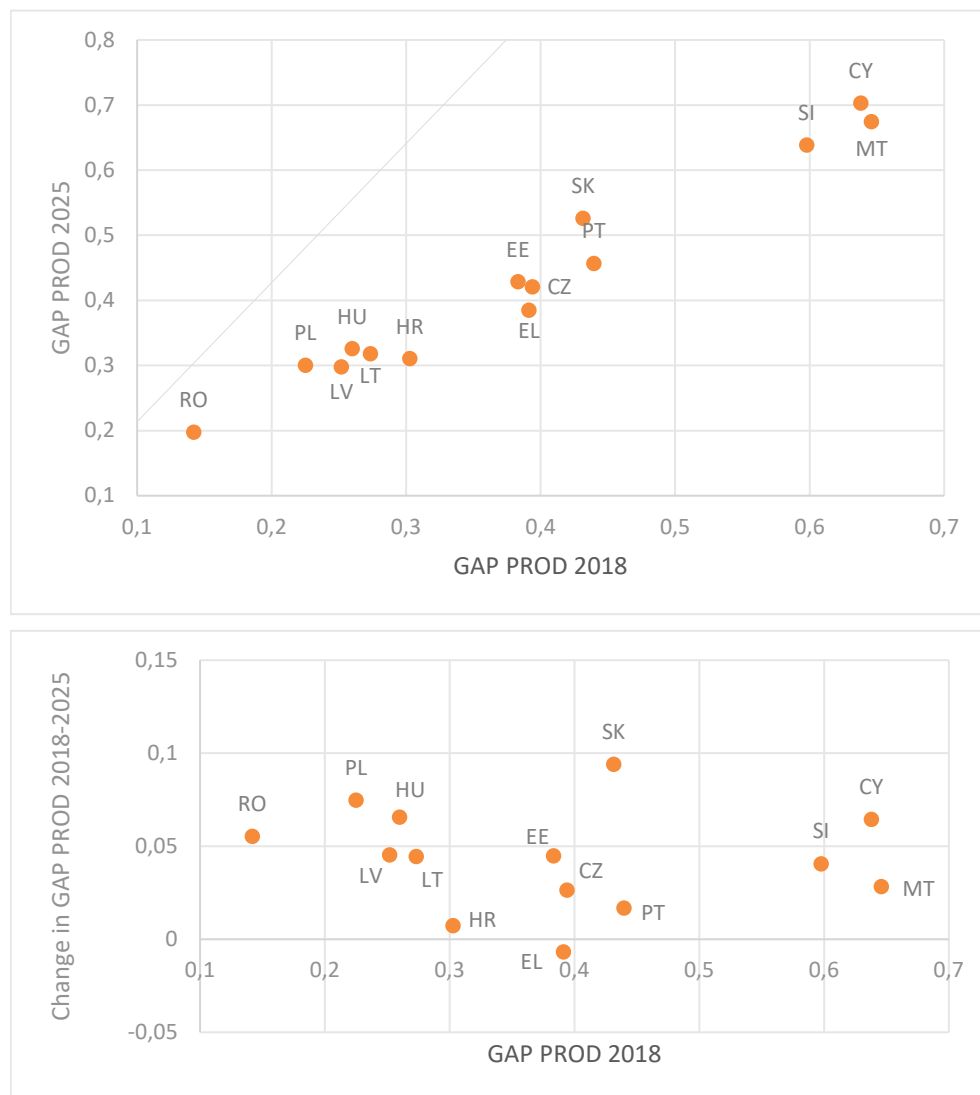
Among the components in EFF, the factor of Labour Productivity (4.3.3.) produces different patterns than the factors related to trade patterns. Labour productivity has a high variance within the EU and a substantial gap for the 'Widening' countries relative to the EU average, with no catching up. If we focus only on Labour productivity, we see the large gap for the typical bottom 'Widening' countries, RO, BG with very little catching up. At the top are the 'Widening' countries CY, MT, SI, which are quite stable leaders on this dimension, countries which are also good performers on factors shaping innovative capacity (cf supra). An outlier in this respect is EE, which is on many factors and on the overall SII indicator the top performer among the 'Widening' sectors, but on Productivity (and also on the EFF in general) only a moderate performer among 'Widening' countries, with little progress over time.

Figure 20 – 'Widening' Countries gap with EU27 average on EFF (2018–2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

Figure 21 – 'Widening' Countries gap with EU27 average on PROD (2018–2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

Our findings on IMPACT (EFF) are the following:

- The gap of the 'Widening' countries in EFF is smaller than the overall SII gap.
- The gap of the 'Widening' countries in EFF has become somewhat smaller. There is some catching up (beta-convergence).
- There is a higher variance in EFF than overall for SII for the 'Widening' countries.
- Variance and gap is especially high in Labour productivity.
- There is no sigma convergence, even a sigma divergence in the EU27 and among 'Widening' countries.
- The gap in EFF between the TOP3 and the BOTTOM3 among 'Widening' countries increased somewhat, with the especially the TOP improving their score. EE is not in the TOP on EFF/PROD; RO and BG are at the bottom on Productivity, a bit less on the broader EFF measure.

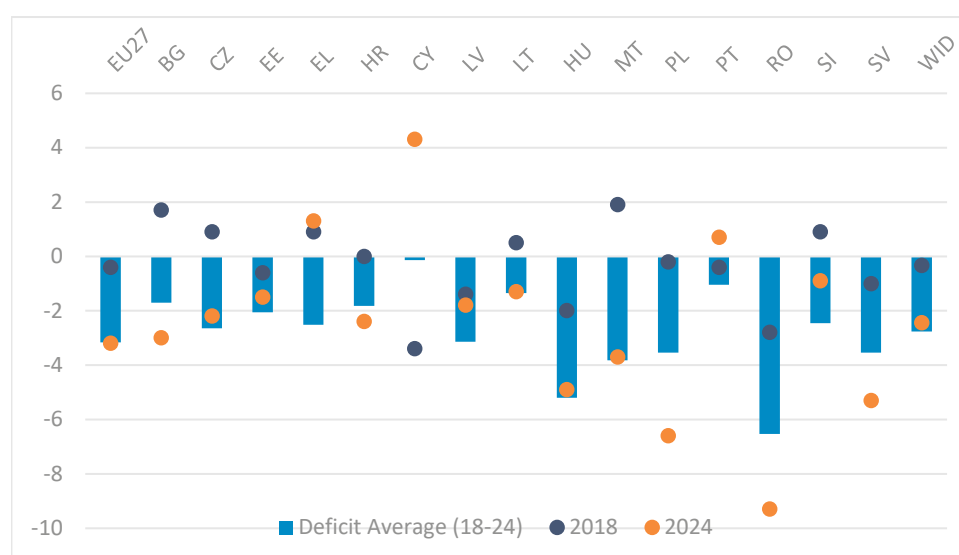
7.3. Context variables for innovation Capacity Building in 'Widening' countries

We finally look at 2 context variables outside the STI/EIS realm, but are important factors to take into account when assessing a country's potential for innovation capacity: the status of its public financing and the rule of law.

7.3.1. Government Deficit

The scope for government support for innovation capacity building requires a sound public finance environment with a room for extra public funding. All 'Widening' countries struggle with government deficit issues in the period considered, 18-24. This holds particularly for Hungary and Romania. 'Widening' countries Exceeding the 3% Deficit Rule (as of April 2025) are Romania, Poland, Slovakia, Hungary, Malta, and Bulgaria.

Figure 22 – 'Widening' Countries Deficit (2018-2025)

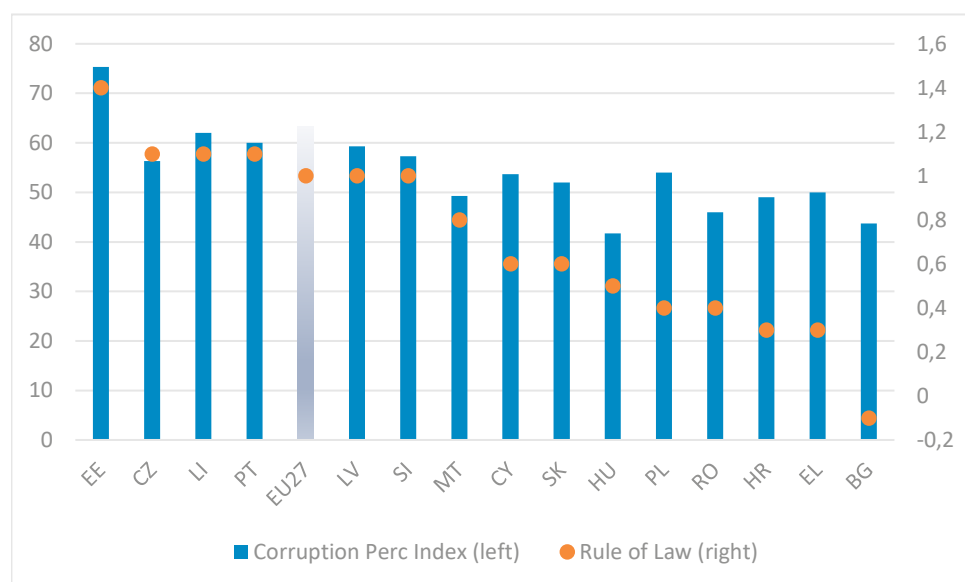


Source: [EUROSTAT](#).

7.3.2. Rule of Law

Although none of the 'Widening' countries score on the Corruption Perception Index significantly lower than the EU27 average, Bulgaria could nevertheless be flagged when looking at the Rule of Law index (World Justice Project).²⁵

Figure 23 – 'Widening' Countries Gap with EU27 average on Rule of Law

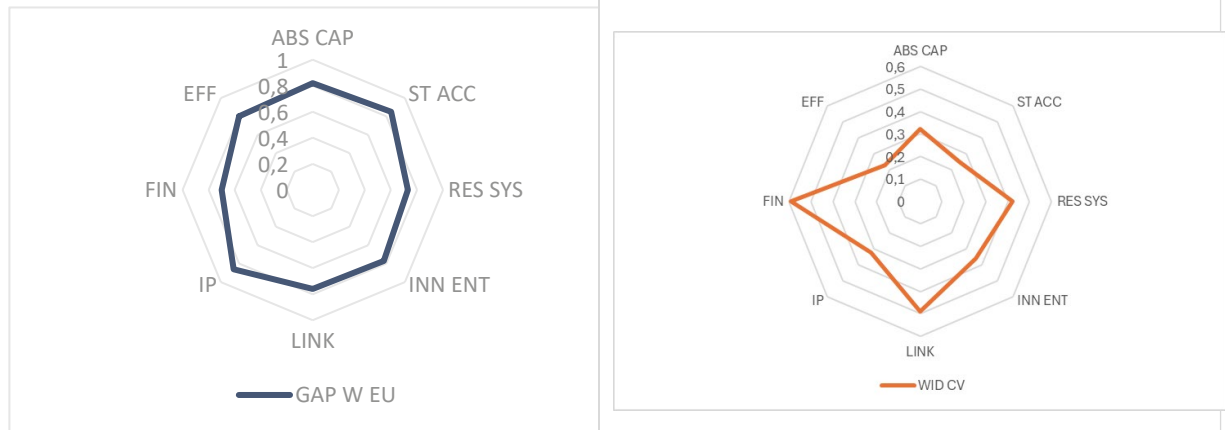


Source: [Corruption Perceptions Index](#) Average 2022-2024 Transparency International and Rule of law (-2.5 to 2.5 best) Average 2021-2023 World Bank: [Worldwide Governance Indicators](#).

²⁵ <https://worldjusticeproject.org/rule-of-law-index/global/2025/Bulgaria/>

8. Comparing gaps and convergence on factors shaping the potential for innovation capacity building

Figure 24 – 'Widening' Countries gap with EU27 average and within CV on factors for Innovation Capacity (2025)



Source: Own calculations based on [EC-JRC EIS2025](#) data.

The gap between 'Widening' countries and the EU27 average holds on all factors for innovation capacity building. The factor with the lowest gap is FIN, but this is also the factor with the highest variance among 'Widening' countries.

Regardless of the heterogeneity among 'Widening' countries, the fact that 'Widening' countries which are more highly developed score better on all conditions for innovation, suggests the need to look at the factors for innovation based growth as a 'system'. Further developing on innovation requires progress on all indicators.

A good illustration of this 'systemic' performance is Estonia. It is the best performing country among 'Widening' countries. This superior scoring of Estonia holds on all the considered factors (the exception being 'effects'). Estonia is a good example of the power of innovation policy. Its stellar performance among transition countries on innovation based growth rides on its systemic Knowledge-Based Estonia (KBE) innovation policy which it started already at an early stage, in 2000, and continued to develop further as Estonia progressed along its innovation development path. Annex 1 describes Estonia's innovation policy in more detail. Estonia's 'systemic' approach to innovation policy is further illustrated by the consistent spending on public R&D which was able to leverage private funding.

9. An assessment of the potential of 'Widening' countries for innovation capacity building

In this final section, we classify all 15 'Widening' countries on their position on the innovation development path and on their *potential* for innovation capacity building. The evaluation of their *potential* for innovation capacity building will be dependent on the initial position of the country.

'Widening' countries need to perform on the prerequisites for innovation capacity building. Two principles guide our analysis in this respect. First, as the analysis has shown the importance of 'systemic' performance, countries need to perform well on all of the prerequisites for innovation capacity building. Second, when defining benchmark levels for scores, we will take into account the position of countries on their innovation development path, ie scores will be compared with the overall 'Widening' benchmark.

Compared to taking the average across all EU27 countries, our procedure reduces the weight of the creative capacity prerequisites, as most of the 'Widening' countries are still far from the technology frontier.

A country is judged as having a *bottleneck* if it scores at the minimum value of all 'Widening' countries (see '-' in Table 12). A *weak spot* is defined as a score below the average for all 'Widening' (see '0' in Table 12).

If we would use as an algorithm (Option 1) that a 'Widening' country with potential for innovative capacity building cannot have more than one bottleneck, not more than two weak spots and cannot accumulate with one bottleneck one other weak spot, the following countries would pass on these criteria: Estonia, Malta, Slovenia, Portugal, and the Czech Republic.

Among these, Slovenia has no weak spots; Estonia has only one weak spot, but this is on the Trade components in Effects, not on Productivity.

Malta also has Effects as a weak spot, but not on Productivity. Malta however also has a bottleneck on Finance. If we use as algorithm (Option 2) that a 'Widening' country with potential for innovative capacity cannot have a bottleneck, Malta would drop out of this group.

Table 12 – Position of 'Widening' countries on the factors shaping the potential to develop and catch up in innovation capacity

	ABS CAP	S&T ACC	RES SYS	INN ENT	LINK	FIN	IP	EFF /PROD
Estonia (EE)	++	++	++	+	++	++	++	0/+
Malta (MT)	++	++	++	+	++	-	++	0/+
Slovenia (SI)	+	++	++	+	++	+	+	+/+
Portugal (PT)	+	++	++	+	++	++	0	0/+
Cyprus (CY)	+	++	++	0	++	-	++	0/+
Lithuania (LT)	+	0	0	0	++	+	0	0/0
Czechia (CZ)	0	+	+	+	+	+	0	+/+
Greece (EL)	0	-	0	+	0	+	0	0/0
Croatia (HR)	0	0	0	+	+	+	0	0/0
Hungary (HU)	0	0	0	0	+	+	0	+/-
Poland (PL)	0	+	-	0	0	+	0	+/-
Slovakia (SK)	+	0	0	0	0	0	0	+/+
Latvia (LV)	+	0	-	-	0	0	0	0/0
Bulgaria (BG)	0	0	-	-	-	-	0	0/-
Romania (RO)	-	-	-	-	-	-	-	+/-

NB ++: score above the EU-27 average (ie no gap); +: score above the 'Widening' average; 0: score below the 'Widening' average; -: extremely low score at the bottom of the 'Widening' group;

Source: Own calculations based on [EC-JRC EIS2025](#) data.

At the bottom of the list, Latvia, Bulgaria and Romania combine several bottlenecks. If we would apply the algorithm described supra, these countries would show as suffering from too many bottlenecks. As they are in addition scoring at the bottom on 'innovation active enterprises', they are too weak to develop a basis for innovation capacity in the near term. They are listed as non-innovation active 'Widening' countries. For these countries, a more useful exercise is to examine whether they are sufficiently strong on absorptive capacity and S&T access, to progress on their trajectory towards innovation-active status. This is the case for Latvia and Bulgaria, but not for Romania.

All other countries are already sufficiently innovation-active, but face on their path towards developing further their innovation capacity, catching up with the EU average, some serious challenges, a combination of a bottleneck challenge and/or several other weak spots. Lithuania, Croatia, Hungary and Slovakia have no bottleneck challenge, but combine several weak spots. Greece, Poland and Cyprus combine several weak spots with a bottleneck challenge: for Greece, this is S&T access, for Poland its Research System and for Cyprus, access to Finance.

Applying this algorithm leads to the classification of 'Widening' countries in the following 5 categories of Potential for Innovation Capacity (PIC).

Table 13 – Assessing 'Widening' countries on their potential to develop and catch up in innovation capacity

	SIZE/DIR SII GAP	INN ACTIVE	PIC
Estonia (EE)	I	1	5
Malta (MT)	I	1	4/5
Slovenia (SI)	I	1	5
Portugal (PT)	I	1	5
Cyprus (CY)	I	1	3
Lithuania (LT)	II	1	4
Czechia (CZ)	II	1	5
Greece (EL)	II	1	3
Croatia (HR)	II	1	4
Hungary (HU)	II	1	4
Poland (PL)	II	1	3
Slovakia (SK)	II	1	4
Latvia (LV)	III	0	2
Bulgaria (BG)	III	0	2
Romania (RO)	III	0	1

Source: Own calculations based on [EC-JRC EIS2025](#) data

10. Participation of 'Widening' countries in the EU's Framework Programme and their Potential for Innovation Capacity

Having constructed a classification of 'Widening' countries on their position on innovation capacity and on their potential for innovation capacity building and catching up, we match this classification with their patterns of participation in Horizon Europe.

For success rates in HEU (see also Table 2), all 'Widening' countries score below the EU27 average, warranting their status as 'Widening'. Among the 'Widening' countries, there is no clear correlation between their PIC status and their HEU success rate.

But if we look at the application behavior (relative to the population of S&E graduates) there is a clear correlation with their PIC score (see Table 14 below). When looking among 'Widening' countries, countries with a higher PIC have on average a higher tendency to apply, taking into account their relevant size. Countries at the bottom of the PIC ranking, with the exception of Latvia, have a lower tendency to apply. This holds also for the size of HEU funding received related to the country's GERD. This suggests that inducing 'Widening' countries to participate in HEU goes hand in hand with a stronger PIC.

Romania, Poland and Hungary are 'Widening' countries which have the lowest application rates (relative to the population of S&E graduates). But at the same time they also have the lowest success rates in HEU, culminating in the lowest HEU funding received per GERD. For Romania, this is correlated with its low scoring on innovation capacity. Also Bulgaria, the other low PIC scoring 'Widening' country, scores at the bottom of the 'Widening' countries for applications, has below average success rates, culminating in bottom performance in HEU funding received per GERD. For Poland and Hungary, this is related to their several weak spots on PIC, among which the quality of their Research Systems and Innovative Enterprises. All these countries are facing public budget deficit problems, exceeding the 3% target, making it difficult to mobilise national funding for improving their potential for innovation capacity building. Particularly Bulgaria and Romania will rely quite heavily on HEU participation and funding, to unlock their access to finance as bottleneck for innovation capacity building.

Latvia also has a low score on potential for innovation capacity building with several bottlenecks (quality of its research system and innovation active enterprises) and several weak spots. Nevertheless, it has an above average application behavior, an above average success rate, culminating in an above average HEU funding received per GERD. Latvia does not exceed the 3% budget deficit target. Lifting Latvia's potential for innovation capacity building through "Widening" requires thinking beyond increasing Latvia's participation in HEU.

Countries at the top of the PIC ranking can gradually be phased out of the 'Widening' programme, conditional on them being able to sustain their growth trend in Innovation Capacity Building and conditional on addressing any small weak points they still might have.

The analysis illustrates that addressing the 'Widening' Participation and Spreading Excellence' to unleash the innovation capacity of lagging countries to catch up, is a complex challenge that requires more than simply addressing participation in EU's Framework Programmes. It requires taking into account the heterogeneity among 'Widening' countries in their position and potential for innovation capacity building.

Table 14 – Matching 'Widening' countries Position on Innovation Capacity with their participation in Horizon Europe

	PIC	HEU SR	HEU APPpS&E	HEU INVpGERD
Estonia (EE)	5	+	++	++
Malta (MT)	4/5	+	++	++
Slovenia (SI)	5	+	++	++
Portugal (PT)	5	0	+	+
Cyprus (CY)	3	0	++	++
Lithuania (LT)	4	+	0	0
Czechia (CZ)	5	+	0	0
Greece (EL)	3	+	++	+
Croatia (HR)	4	0	+	0
Hungary (HU)	4	+	-	-
Poland (PL)	3	0	-	-
Slovakia (SK)	4	+	-	-
Latvia (LV)	2	+	+	+
Bulgaria (BG)	2	0	0	-
Romania (RO)	1	0	-	-

NB: 0: below 'Widening' average; -: at the bottom of the 'Widening' average; +: above 'Widening' average for HEU SR; above EU average for HEU APP per S&E and HEU INV per GERD; ++ above 'Widening' and EU average; Grey shaded are the 'Widening' countries exceeding the 3% deficit Rule in 2025;

Source: Own calculations based on DG-JRC EIS2025 and DG-RTD (2025), Interim Evaluation of the Horizon Europe Framework Programme for Research and Innovation (2021 - 2024)²⁶.

²⁶ https://research-and-innovation.ec.europa.eu/strategy/support-policy-making/shaping-eu-research-and-innovation-policy/monitoring-evaluation-and-impact-assessment-horizon-europe_en

11. 'Widening' index options

Based on Tables 12, 13 and 14, several options for 'Widening' countries support in the next FP could be considered.

For all three options and regardless number of selected countries, it is very important to note that future 'Widening' measures should not only on support participation in FP in general, but specifically target those programmes/pillars tackling a country's weaknesses with the best scope for PIC improvement.

'Widening' index option 1: Strict scenario

The first scenario omits the top five best-performing 'Widening' countries. Using an algorithm to select 'Widening' countries with potential for innovative capacity building according to the criteria that: they do not have more than one bottleneck, have no more than two weak spots and do not accumulate one bottleneck with one other weak spot; the following countries would pass: **Czechia, Estonia, Malta, Portugal** and **Slovenia**. Among these, Slovenia has no weak spots; Estonia has only one weak spot (on the trade components under 'effects', not on productivity). These countries can be gradually phased out of the 'Widening' support, on the condition that they will sustain their innovation capacity building growth trend and that they will address any persisting weak points.

'Widening' index option 2: Moderate scenario

The second scenario loosens the above selection criteria. Using a more conservative algorithm to select 'Widening' countries with potential for innovative capacity that do not have a bottleneck, Malta would drop out of the top group, as it has a bottleneck on finance. Only **Czechia, Estonia, Portugal** and **Slovenia** would therefore remain in the best-performing group. As in Option 1, these countries can be gradually phased out of 'Widening' support.

'Widening' index option 3: Inclusive scenario

The third scenario would retain all 'Widening' countries within the 'Widening' support. All 'Widening' countries still score below the EU-27 average, when cross-referencing their PIC ranking with the success rate of their HEU funding applications, confirming their status as 'Widening'. While there is no clear correlation between 'Widening' countries and their PIC status, ranking currently corresponds (except for **Greece** and **Latvia**) to the Horizon Europe funding received per their gross domestic expenditure on research and development (GERD). Continued support could use PIC ranking to tailor widening instruments according to bottlenecks and weak spots for identified Member States.

12. Concluding recommendations

Based on our analysis of the evidence on the innovation development and HEU involvement of 'Widening' countries, we propose the following recommendations, very much in line with the draft proposal on (European Commission, 4 July 2025 communication).

- Call for differentiating among 'Widening' along their position in the innovation capacity development path and depending on their capacity to further develop (see PIC).
- This differentiating should be translated into 'Widening' instruments tailored to these subcategories.
- 'Widening' instruments could even be tailored made for individual countries depending on their weak spots (eg international connectivity; quality of HEI institutes, digital skills...).
- Focus not only on supporting participation in FP in general, but on supporting participation in those programmes/pillars tackling a country's weak spots and which have the best scope for improving a country's PICs.
- Select and evaluate 'Widening' programmes and participation in FP into effectiveness on PIC.
- Select and evaluate 'Widening' programmes and participation in FP on leveraging these into national public and private capacity building.
- Remove bottlenecks for 'Widening' countries (especially those in 3 % deficit) to activate national funding for interventions that are PIC cleared.
- Remove bottlenecks for using other EU funding (e.g. Structural Funds) for funding interventions that are PIC cleared.
- Do regular country profiling on PIC (eg see OECD Innovation System profiling of countries).
- Monitor, evaluate and remedy 'Widening' countries usage of national and EU instruments, using e.g. the Semester toolbox.

13. Caveats and suggestions for further analysis

The methodology to classify countries into 'Widening' status and within 'Widening' status into their PIC status relies on EU-JRC Innovation Scoreboard data, which are tested and widely used indicators in innovation policy circles. Nevertheless, indicators could be improved and added to better reflect the factors that shape innovation capacity building.

- E.g. on linkages, patent and publication citation information could be used to finetune the link between science and technology.
- E.g. on innovation active enterprises, more information on new and young firms, including science-based start-ups could measure deeper the scope for business dynamism.
- E.g. on international connectedness, more information on international mobility of researchers and entrepreneurs to capture brain circulation, brain drain, brain gain.

The proposed methodology composes factors by using multiple indicators as input, and thus reduces the sensitivity of the classification outcome to individual factors and outliers, increasing the robustness of the classification.

Also, the systemic nature of innovation capacity building, requiring a simultaneous presence of prerequisites, lead to a rather robust classification, as performance correlates across factors. Bottom performers score typically low on all indicators, top performers high.

Nevertheless, robustness checks with various scenarios of different weights to individual components within factors and to different weights of factors in the overall PIC score, should be on the radar, if only to show the sensitivity of the results to each component.

More analysis is needed to match the size and format of 'Widening' support to the various PIC categories.

More analysis is needed to evaluate how 'Widening' support contributes to

- improvements in PIC scores for recipients,
- reduce gaps between 'Widening' countries and the EU27 average,
- improve EU27 cohesion and to improve EU27's overall innovation capacity.

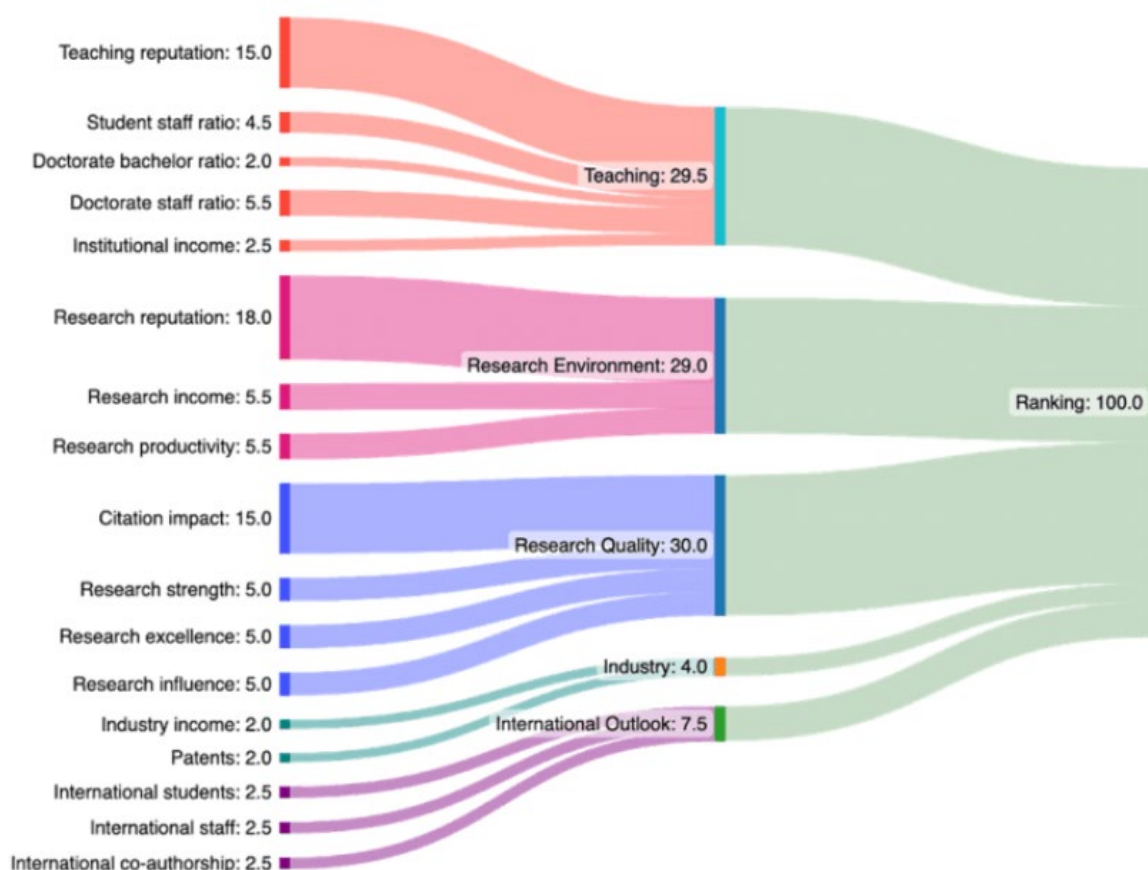
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Annex 1: Times Higher Education Ranking

The *Times Higher Education World University Rankings* ranks research-intensive universities across all their core missions: teaching, research, knowledge transfer and international outlook.²⁷ 18 performance indicators provide a comprehensive and balanced comparison.

The performance indicators are grouped into five areas: Teaching (the learning environment); Research environment (volume, income and reputation); Research quality (citation impact, research strength, research excellence and research influence); International outlook (staff, students and research); and Industry (income and patents). Each indicator gets a weight within its area and each area gets a weight in the final score each institution receives.



To calculate the score of a country on its THE ranking, we calculate the share of its HEI institutes in each ranking class and then multiplies this share with the score allocated by THE to each of these categories.

²⁷ <https://www.timeshighereducation.com/world-university-rankings/world-university-rankings-2025-methodology>

THE rank category	Score
100-200	65.5
200-250	58.6
250-300	55.7
300-350	53.6
350-400	51
400-500	49.2
500-600	45.9
600-800	43.2
800-1000	38.1
1000-1200	34.4
1200-1500	30.6
>1500	25.1

For the institutes in the <100 category, they each get an individual score. For these institutes we take the average of the individual scores of each of the HEIs in this category.

Annex 2: Estonia's innovation policy (based on Veugelers & Schweiger (2016))

Estonia's innovation policy formally began in 2000 with discussions regarding the first Knowledge-Based Estonia (KBE) strategy, which covered the period 2002–06. This strategy drew on the experiences of Finland and Sweden (see, e.g., OECD 2010; Polt et al. 2007), taking account of specific development opportunities, the existing research potential and the country's economic structure, as well as other Estonian development strategies, most notably its education policy strategy 'Learning Estonia'. The two main objectives were updating Estonia's knowledge pool and increasing the competitiveness of its companies. The three key areas for Estonian research, development and innovation (RDI) were (i) user-friendly information technology and the development of an information society; (ii) biomedicine; and (iii) material technology (see KBE 2002–06).

In order to achieve these objectives, the KBE strategy established a set of measures spanning four key areas: financing of R&D, development of human capital, increasing the effectiveness of RDI systems and international cooperation. These measures sought to increase gross domestic expenditure on R&D (GERD) to 1.5 per cent of GDP by 2006. They also aimed to rebalance expenditure on research versus development, seeking to shift the breakdown between the two from 90:10 to 60:40 by 2006.

To increase the effectiveness of its RDI system, Estonia adopted location-based policies, creating science parks and regional business incubators. Lastly, Estonia used international cooperation not only as a means of attracting foreign knowledge and technology, but also as a way of building research teams with critical mass and avoiding 'brain drain'.

The KBE strategy for the period 2002–06 has been followed by similar strategies for the periods 2007–13 and 2014–20. A number of governmental and independent bodies have conducted assessments looking at the progress made under the first two strategies. Each strategy has taken account of the experience and expert recommendations resulting from the preceding period and set more ambitious objectives, with targets increasing in number and scope. The key areas have been adjusted over time, but the overall priorities have not. The focus continues to be on ICT, health technology and services, and more efficient use of resources.

Private-sector involvement helped to strengthen publicly funded education and training programmes. The Estonian Association of Information Technology and Telecommunications (EAITT), an industry association, played a leading role in the development of clusters and the design of vocational and university education programmes.

The EU's Horizon Europe framework programme promotes European innovation. The current Horizon Europe programme uses 'Widening' measures to target Member States, support their capacity to innovate, close gaps between EU countries and move towards performance convergence. This study provides a framework and associated indicator options for establishing a systemic 'Widening 2.0' methodology which could prove instrumental in discussing post-Horizon Europe 'widening' support measures. It is based on assessment of research and innovation indicators of targeted 'widening' countries, looking at the economic and research impact that Horizon Europe may have had between 2018 and 2025.

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